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Single Domain Antibody Inhibitors Target the Coiled Coil Arms of the *Bacillus subtilis* SMC complex

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eLife Assessment

This **important** study introduces an innovative synthetic nanobody approach to probe the function of the bacterial SMC complex. The authors provide **convincing** evidence that these nanobodies target the coiled-coil region of the SMC subunit and demonstrate that this region is critical for SMC function in vivo. Overall, the work is significant for the fields of genome organization, SMC protein biology, synthetic biology, and bacterial cell biology.

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Abstract

Synthetic nanobodies—also called sybodies—have proven valuable for stabilizing conformations of purified proteins, advancing structural and functional studies for example of transmembrane protein complexes. However, their utility in modulating protein function in living cells has remained less well explored. Structural Maintenance of Chromosomes (SMC) complexes facilitate chromosome organization by DNA loop extrusion, a fundamental process in all domains of life. In this study, we target the bacterial SMC complex, Smc-ScpAB, in *Bacillus subtilis* with synthetic nanobodies, with the aim to identify key functional regions of the protein complex in a largely unbiased manner. We first isolate sybodies that specifically bind purified Smc-ScpAB and then express them in *B. subtilis* to select binders capable of disrupting Smc-ScpAB function, leading to chromosome segregation defects and cell death. Mapping and biochemical characterization show that the fourteen disruptive sybodies belong to one of three library designs, target the Smc subunit near the same coiled coil arm interface and modulate its ATPase activity in two principal ways, highlighting the mid-region of the Smc coiled coil as critical feature of the DNA folding process. These findings underscore the potential of sybodies—and, by extension, designed binders—as versatile tools for probing dynamic protein function in living cells, with potentially broad applications in cell and synthetic biology.

Introduction

Rapid and specific interference with protein activity and dynamics in living cells is essential for studying and understanding biological mechanisms. However, current approaches—primarily based on small-molecule inhibitors—remain laborious, time-consuming, and costly. The advent of conformation-specific synthetic nanobody (sybody) selection (Zimmermann et al., 2018 [\[link\]](#)), and more recently, computational design of protein binders (Pacesa et al., 2025 [\[link\]](#)), offer unique opportunities for protein-based inhibition and modulation of cellular targets. Here, we generate sybodies against the SMC complex in *B. subtilis* to efficiently inhibit its function within cells.

Structural Maintenance of Chromosomes (SMC) complexes are multi-subunit, ring-shaped ATP-hydrolyzing DNA motors that structure chromosomal DNA by loop extrusion. They are essential for chromosome organization and segregation, gene expression, DNA repair, and defense against

non-self DNA across all domains of life loop (Yatskevich, Rhodes and Nasmyth, 2019 [↗](#)). Each SMC protein (Smc in *B. subtilis*, *bsuSmc*) is a long polypeptide with its N- and C-termini folding together to form a globular ATP binding cassette (ABC) “head” domain. The head is connected to a “hinge” dimerization domain via a ~50 nm antiparallel coiled coil “arm”, creating an elongated dimer (Figure 1Aii [↗](#) and 1Aiii [↗](#)). ATP binding promotes head engagement, bringing the heads together to form a functional ATPase. Non-SMC subunits, a kleisin (ScpA in *B. subtilis*) and a dimer of KITE (ScpB in *B. subtilis*) or two HAWK proteins, bridge the heads, together forming a ring around DNA (Figure 1Aiii [↗](#)). In *B. subtilis*, Smc-ScpAB is recruited to the origin of replication (*oriC*) region on the chromosome by ParB, a DNA-binding protein that recognizes centromere-like *parS* sequences (Figure 2A [↗](#)). Once loaded, Smc-ScpAB translocates at a rate of ~1 kb/s onto flanking DNA, aligning the two chromosome arms and individualizing nascent sister chromosomes (Gruber et al., 2014 [↗](#); Bürmann and Gruber, 2015 [↗](#); Wang et al., 2017 [↗](#)), likely bypassing obstacles on the chromosome through the SMC hinge gate (Liu et al., 2025 [↗](#)). Null mutants of *smc*, *scpA*, or *scpB* fail to segregate chromosomes properly and lose viability under conditions promoting rapid growth (Sullivan, Marquis and Rudner, 2009 [↗](#); Gruber et al., 2014 [↗](#); Wang et al., 2014 [↗](#)).

Several models have been proposed for the mechanism of DNA loop extrusion by SMC complexes, including the “segment capture” model, in which DNA segments are transiently trapped between the SMC arms in partially open states and fused into larger loops through iterative ATP-driven cycles (Figure 1Aiii [↗](#) and S1B [↗](#)) (Minnen et al., 2016 [↗](#); Diebold-Durand et al., 2017 [↗](#); Marko et al., 2019 [↗](#)). However, detailed structural understanding and experimental testing *in vivo* are lacking.

Antibodies can block biochemical reactions by stabilizing reaction intermediates. Synthetic single-domain antibodies called sybodies are small, robust antigen-binding proteins that were engineered based on three camelid nanobody structures (Zimmermann et al., 2018 [↗](#)). They are selected from synthetic libraries that encode a wide diversity of binding surfaces and epitope shapes based on the length and geometry of their CDR3 loop, allowing strong binding of diverse antigen surfaces (Figure 1Ai [↗](#)). Their small size (~15 kDa), stability, and ability to bind transient epitopes make them ideal tools for targeting specific states of biomolecules (Rasmussen et al., 2011 [↗](#)). Selection is carried out using purified and immobilized Smc-ScpAB by ribosome display and phage display, typically followed by an ELISA-based screening (Zimmermann et al., 2018 [↗](#)).

Here, we demonstrate that sybodies can be used to interfere with the function of *B. subtilis* Smc-ScpAB *in vivo*. We first isolate binders that specifically target purified Smc-ScpAB *in vitro* and then select those that eliminate Smc function when expressed in *B. subtilis*. Fourteen sybodies were found to disrupt chromosome segregation, mimicking *smc* deletion phenotypes. Biochemical assays confirmed that selected sybodies alter the ATP hydrolysis rate of Smc-ScpAB and its stimulation by DNA addition, likely stabilizing otherwise rare intermediates of the conformational cycle. Mapping experiments using chimeric Smc constructs revealed that most binders, unexpectedly, target the coiled coil, specifically in a defined region near the Smc 4N arm-to-arm contact, highlighting the potential of sybodies and designed binders, as genetic tools to identify essential functional domains and underscoring the importance of coiled coil dynamics for Smc function.

Results

Generation of *bsuSmc*-ScpAB-specific sybodies

To isolate sybodies that impede the function of *B. subtilis* Smc-ScpAB complex, we first performed *in vitro* selections with purified Smc-ScpAB under conditions favoring an ATP-bound state, using 40 bp duplex DNA, and an ATPase-deficient mutant (E1118Q) of *bsuSmc* biotinylated at the hinge at residue R643C for immobilization (Figure 1A [↗](#) and S1A [↗](#)) (Hirano and Hirano, 2004 [↗](#); Bürmann et al., 2017 [↗](#); Zimmermann et al., 2018 [↗](#); Zimmermann et al., 2020 [↗](#)). Starting from three synthetic sybody libraries encoding distinct epitope-binding geometries (denoted as concave, loop, convex, respectively, Figure 1A [↗](#)), we performed one round of ribosome display followed by two

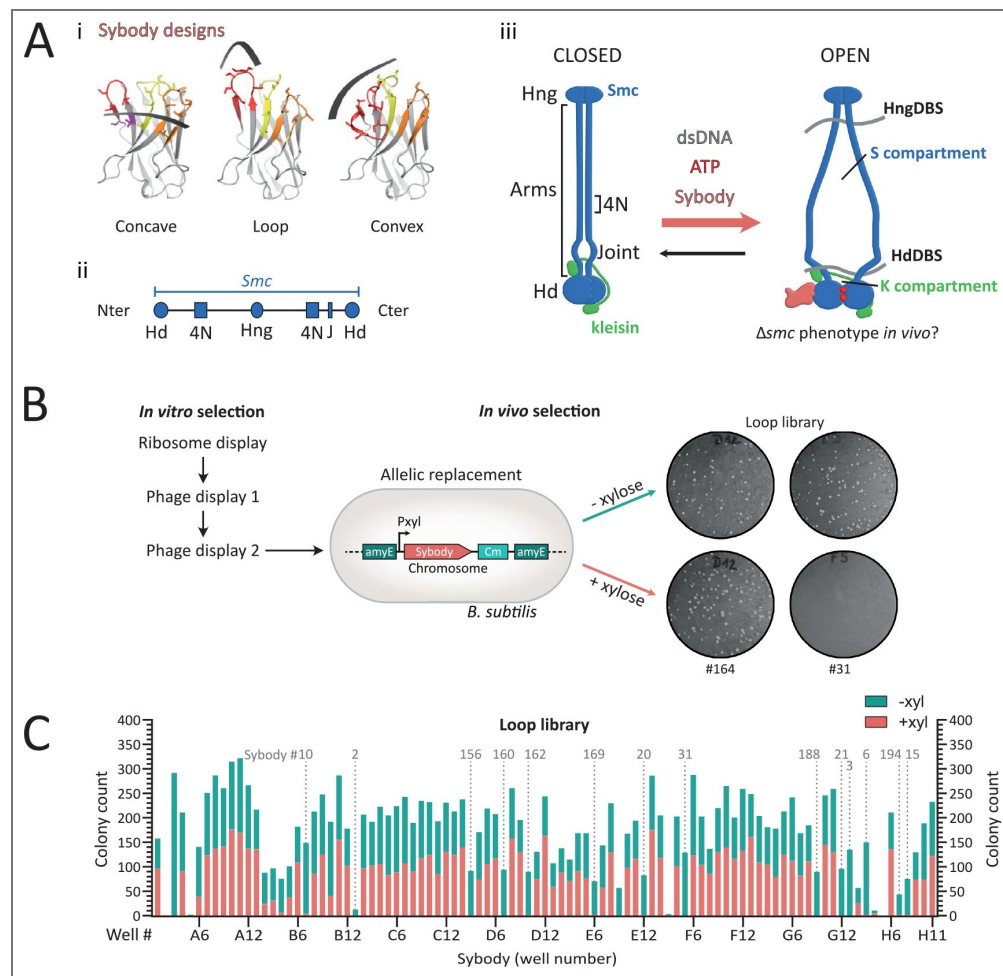


Figure 1. Selection of sybodies targeting Smc-ScpAB in *B. subtilis*.

(A) (i) Structural design and randomization scheme of the three synthetic sybody libraries: concave, loop, and convex. Complementarity-determining regions (CDRs) 1, 2, and 3 are shown in yellow, orange, and red, respectively; randomized residues are shown as sticks. Adapted from (Zimmermann et al., 2018). **(ii)** SMC domain organization. The N- and C-terminal regions encode half an ATPase head, while the middle domain constitutes the hinge dimerization domain. The elongated sequence in-between encodes for long helical sequences that will form an antiparallel intramolecular coiled coil. Hd: head, Hng: hinge, 4N: 4N arm-to-arm contact, J: joint. **(iii)** SMC complexes harbor juxtaposed arms in their resting state. Upon ATP binding at the heads, the heads dimerize (head engagement), leading to a conformational change of the entire complex in which the arms open. This state is thought to fit a loop of DNA (not shown) in the S compartment formed by the SMC arms, that will be pushed towards the K compartment formed by the heads and kleisin, after ADP release. We hypothesized that SMC-blocking sybodies can stabilize distinct conformation of the *bsuSmc-ScpAB* complex promoted by ATP hydrolysis and DNA binding, potentially inducing a Δsmc -like phenotype *in vivo* and enabling biochemical characterization of trapped conformations *in vitro*. Blue: SMC dimer, Green: kleisin ScpA. The ScpB KITE subunits are not represented for the sake of simplicity. HngDBS: Hinge DNA-binding site, HdDBS: Head DNA-binding site. **(B)** Framework for sybody selection. *In vitro* selection starts with $\sim 10^{12}$ sybody variants per library subjected to ribosome display for pre-enrichment, followed by two rounds of phage display. For *in vivo* selection, 95 randomly selected sybody genes were integrated into the *B. subtilis* chromosome under the control of a xylose-inducible promoter by allelic replacement at the *amyE* locus. Growth defects on rich medium were tested on ONA agar plates with or without 0.5% xylose. Example shown: Sb164 (loop library) did not affect growth, whereas Nb31 impaired growth upon induction, suggesting interference with *bsuSmc* function. **(C)** Growth assay results for the 95 *B. subtilis* strains expressing individual xylose-inducible sybodies of the loop library. Bars show colony counts “without” (green) on top of “with” (pink) xylose. Strains are ordered by their original position in the 96-well plate. Fourteen sybodies consistently impaired colony formation under inducing conditions (marked by dotted lines). Sybody numbers indicated above the plots correspond to selected candidates used in subsequent experiments, numbering according to order of first use. Notably, the strain corresponding to the E09 sybody (Sb018), showed an absence of colonies upon sybody induction. However, this sybody candidate gave intermediate phenotypes in later experiments, which is why it was excluded from detailed analysis.

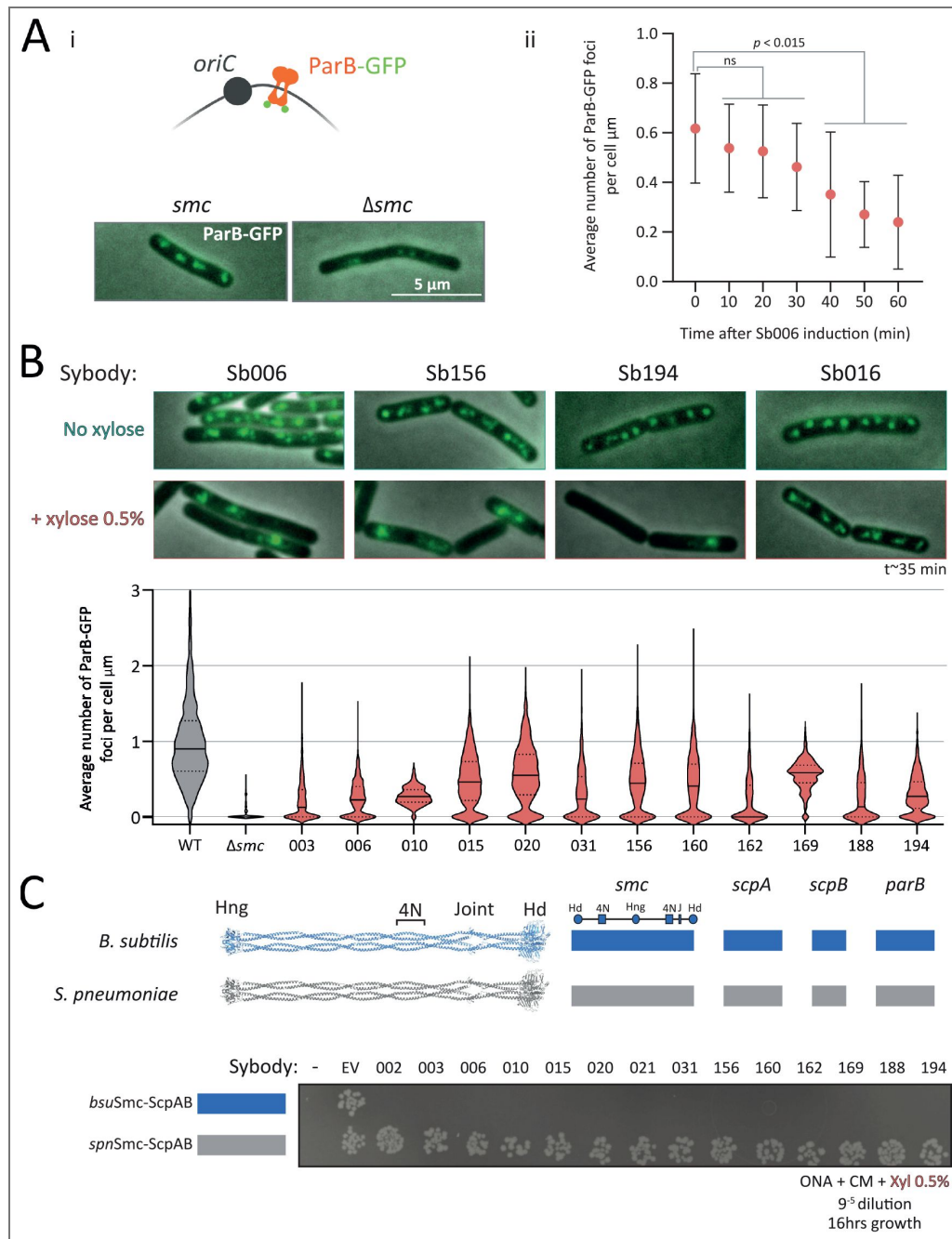


Figure 2. Sybody-induced chromosome segregation defects visualized by ParB-GFP imaging.

(A) i, Top: Schematic illustrating ParB-GFP binding near *oriC*, enabling visualization of origin positioning. Bottom: Representative images showing ParB-GFP foci in wild-type and Δsmc *B. subtilis*. *Smc* WT cells typically display 2–4 foci per cell, whereas Δsmc cells exhibit reduced foci numbers. ii, Number of ParB-GFP foci per μm of cell length in a strain carrying inducible sybody Sb006. A significant decrease in foci is detected from 40 minutes post-induction ($p_{t0-10} = 0.9674$; $p_{t0-20} = 0.9033$; $p_{t0-30} = 0.3744$; $p_{t0-40} = 0.0145$; $p_{t0-50} = 0.0004$; $p_{t0-60} < 0.001$). Based on this, a standardized induction time of ~35 minutes was used in all subsequent experiments. (B) ParB-GFP foci density (foci/ μm) in WT, Δsmc , and sybody-expressing strains after 35 minutes of xylose induction. Violin plots show distribution per condition; solid lines denote the mean, and dotted lines indicate quartiles. (C) Spot assay to assess colony formation of *B. subtilis* strains *bsu* or *spn* variants of *Smc*-ScpAB and ParB. Top: Schematic shows gene origins (blue: *B. subtilis*; grey: *S. pneumoniae*). The leftmost column corresponds to the parental *B. subtilis* strain 1A700 carrying the *S. pneumoniae* genes without chloramphenicol resistance (non-growing). The next spots represent the same strains carrying no sybody gene but the Cm resistance (EV for empty vector). Remaining spots are sybody-expressing strains; sybody numbers are indicated above. Cells were grown for 16 h at 37 °C on ONA supplemented with 0.5% xylose and chloramphenicol. Hd: head, Hng: hinge, 4N: 4N arm-to-arm contact, J: joint.

rounds of phage display, thereby enriching sybodies that bind to ATP- and DNA-bound Smc-ScpAB (Table 1 [↗](#)). From each library, 95 sybody-expressing *E. coli* clones were randomly chosen for *in vivo* characterization.

Identification of *bsu*Smc-ScpAB-blocking sybodies through phenotypic screening in *B. subtilis*

Despite containing a conserved disulfide bond, some sybodies can be expressed inside cells and bind their targets in the reducing environment of the cytoplasm, likely owing to their robust folding (Deneka et al., 2021 [↗](#)). To assess whether the selected sybodies interfere with Smc function *in vivo*, each of the 285 sybody sequences was cloned under a xylose-inducible promoter (P_{xy}) into an *E. coli*–*B. subtilis* shuttle vector and integrated at the *amyE* locus of the wild-type 1A700 strain (Figure 1B [↗](#)) (Diebold-Durand, Bürmann and Gruber, 2019 [↗](#)). Fourteen sybodies from the loop library failed to yield transformants in the presence (but not absence) of xylose, indicating cell lethality due to sybody inhibition of Smc-ScpAB (Figure 1C [↗](#), S2A). These Smc-disruptive sybodies (denoted as Sb002, 003, 006, 010, 015, 020, 021, 031, 156, 160, 162, 169, 188, and 194, respectively) harbor fourteen distinct sequences, showing the absence of duplicates (Table S1 [↗](#)). Intriguingly, no such disruptive sybodies were isolated from the concave and convex libraries (Figure S2B [↗](#)), suggesting that the CDR3 geometry as present in the loop library is particularly effective at targeting *bsu*Smc-ScpAB. Alternatively, the other libraries did not produce good binders or these sybodies were not stably expressed in *B. subtilis*. Notably, tendencies of preferential isolation of binders from one of the three libraries have previously been observed, although at milder levels (Zimmermann et al., 2018 [↗](#); Hutter et al., 2019 [↗](#); Deneka et al., 2021 [↗](#)).

Selected sybodies target *bsu*Smc-ScpAB function *in vivo*

To determine whether the fourteen sybodies indeed impair chromosome segregation, we used fluorescence microscopy to monitor *oriC* positioning in a *B. subtilis* strain expressing a ParB-GFP fusion protein together with a P_{xy} -inducible sybody gene. In these strains, ParB-GFP marks the replication origin region by binding to *parS* sites near *oriC* (Figure 2Ai [↗](#)). As expected, the Δsmc strain displayed fewer ParB-GFP foci and elongated cells, consistent with delayed *oriC* separation (Figure 2Ai [↗](#)) (Table S2 [↗](#)) (Gruber et al., 2014 [↗](#); Wang et al., 2014 [↗](#)). A time course experiment using sybody Sb006 revealed chromosome organization defects as early as 30 minutes post induction, with a significant reduction in ParB-GFP foci per μm cell length at 40 minutes ($p = 0.0145$) (Figure 2Aii [↗](#)). Subsequent imaging was performed ~ 35 minutes after induction of sybody expression.

All disruptive sybodies reduced ParB-GFP foci density compared to the control (1.03 foci/ μm), with values ranging from 0.56 foci/ μm (Sb020) to 0.22 foci/ μm (Sb003). These defects were all robustly detected, while being less severe than the Δsmc mutant (~ 0.05 foci/ μm), conceivably due to the short induction time or incomplete inhibition. Milder defects were also observed without induction, likely attributed to leaky expression from the P_{xy} promoter (Figure S3A [↗](#)). Sybody expression also caused cell elongation and growth delays, both hallmarks of impaired Smc activity, consistent with chromosome segregation defects that delay cell division (Figure S3B–C [↗](#)). Altogether, these results show that all selected sybodies induce Δsmc -like phenotypes likely by interfering with *bsu*Smc-ScpAB activity *in vivo*.

Selected sybodies specifically target *B. subtilis* Smc-ScpAB

To test whether sybodies specifically target *B. subtilis* Smc-ScpAB, we assessed their effects in a strain expressing *Streptococcus pneumoniae* (*Spn*) Smc-ScpAB and ParB proteins in place of the endogenous *B. subtilis* proteins. The *Spn* sequences can functionally replace the corresponding *Bsu* sequences despite significant sequence divergence ($\sim 38\%$ sequence identity) (Bock et al., 2022 [↗](#)). Strikingly, none of the sybodies impaired growth in this background, even though a cognate ParAB pair is absent in this strain (*S. pneumoniae* lacks *parA*), sensitizing cells to chromosome segregation defects (Figure 2C [↗](#)). These results confirm that the sybodies are specific to *bsu*Smc-ScpAB and that off-target toxicity is not noticeable.

bsuSmc-ScpAB-disrupting sybodies target two distinct coiled coil regions adjacent to the bsuSmc 4N arm-to-arm contact

To map the sybody-binding site on the *bsuSmc-ScpAB* complex, we utilized five chimeric Smc constructs (Smc Chimera 1-5), in which the hinge and progressively longer segments of the adjacent coiled coils were replaced with the corresponding sequences from the *S. pneumoniae* Smc protein (Figure 3A). Chimeric junctions were designed to preserve coiled coil integrity based on available crystal structures, coiled coil predictions by DeepCoil, and AlphaFold2 structural models (Zimmermann et al., 2018; Gabler et al., 2020; Jumper et al., 2021). All chimeric Smc strains retained the native *B. subtilis* *scpA*, *scpB*, and *parB* genes and were viable under conditions promoting fast growth, demonstrating proper protein folding and functioning of chimeric Smc-ScpAB complexes (Figure S4A).

Chimeras 1 and 2, which carry the central hinge and coiled coil sequences of *Spn* origin (residues 363-812 and 339-836, respectively), showed the same sybody-induced growth defects as the wild-type strain, indicating that the sybodies bind further away from the hinge, within the remaining *B. subtilis* portion comprising of the Smc head and the first ~230 coiled coil residues, and ScpAB (Figure 3A). By contrast, Chimera 5 (residues 248-927 of *Spn* origin), supported robust growth even in the presence of the *bsuSmc*-disrupting sybodies, indicating that the region comprising the heads and first ~50 residues of the coiled coil (including the joint) does not comprise the binding site. The binding region for the sybodies is thus located on the coiled coil between residues 248 and 339 (or 836 and 927), demonstrating that all fourteen sybodies bind to a central region of the coiled coil.

Chimera 3 (*spnSmc* residues 318-857) and chimera 4 (*spnSmc* residues 276-899) revealed two distinct sybody responses. Sb003, 031, 156, 160, 162, and 169 impaired growth of both strains, mapping their binding sites to *B. subtilis* residues 248-276 (and 899-927), present in chimeras 1-4 but absent in chimera 5. In contrast, Sb002, 006, 010, 015, 020, 021, 188, and 194 had no impact on the viability of chimera 3 or 4, suggesting they target residues 318-339/836-857 (Figure 3A, 3D).

In sum, these experiments mapped the binding sites of all fourteen disruptive sybodies to one of two ~twenty amino acid segments in the central region of the *bsuSmc* coiled coil (Figure 3D). These regions harbor relatively poorly conserved sequences but flank the 4N arm-arm contact (~at residue 295), previously proposed to serve as an essential structural switch between closed and open conformations (Vazquez Nunez et al., 2021). These results reveal that all isolated sybodies target the same functional element of the Smc arms, and that interfering with this element disrupts Smc-dependent chromosome organization *in vivo*.

Smc-disrupting sybodies affect the ATPase activity in one of two ways

To investigate how sybodies may influence *bsuSmc-ScpAB*, we next measured ATP hydrolysis rates using purified proteins in the presence and absence of DNA. SMC ATPase activity serves as an indirect readout for state transitions, including arm opening and head engagement. In absence of DNA, *bsuSmc-ScpAB* hydrolyzes ATP at ~4 ATP/min/Smc (Vazquez Nunez et al., 2021). DNA binding stimulates ATP hydrolysis to ~9 ATP/min/Smc, presumably by DNA binding at the hinge promoting arm opening and head engagement, creating a more open conformation.

All disruptive sybodies measurably affected the ATP hydrolysis rate of *bsuSmc-ScpAB*. Ten maintained a near-basal ATPase rate even in presence of DNA (Sb002, Sb003, Sb010, Sb031, Sb156, Sb160, Sb162, Sb169, Sb188, and Sb194) (Figure 3B-C and S4B). This suggests that they stabilize the complex in a closed or partially closed conformation, thus hindering DNA-dependent ATPase stimulation. Notably, in this group, Sb002 and Sb194 showed slightly elevated ATPase rates with DNA, hinting at a mildly more flexible conformation.

In contrast, the other sybodies (Sb006, Sb015, Sb020, and Sb021) stimulated ATP hydrolysis in the absence of DNA, with Sb006 reaching levels comparable to DNA-induced activation (~10 ATP/min). This implies that these sybodies stabilize Smc-ScpAB in a more open state promoting head

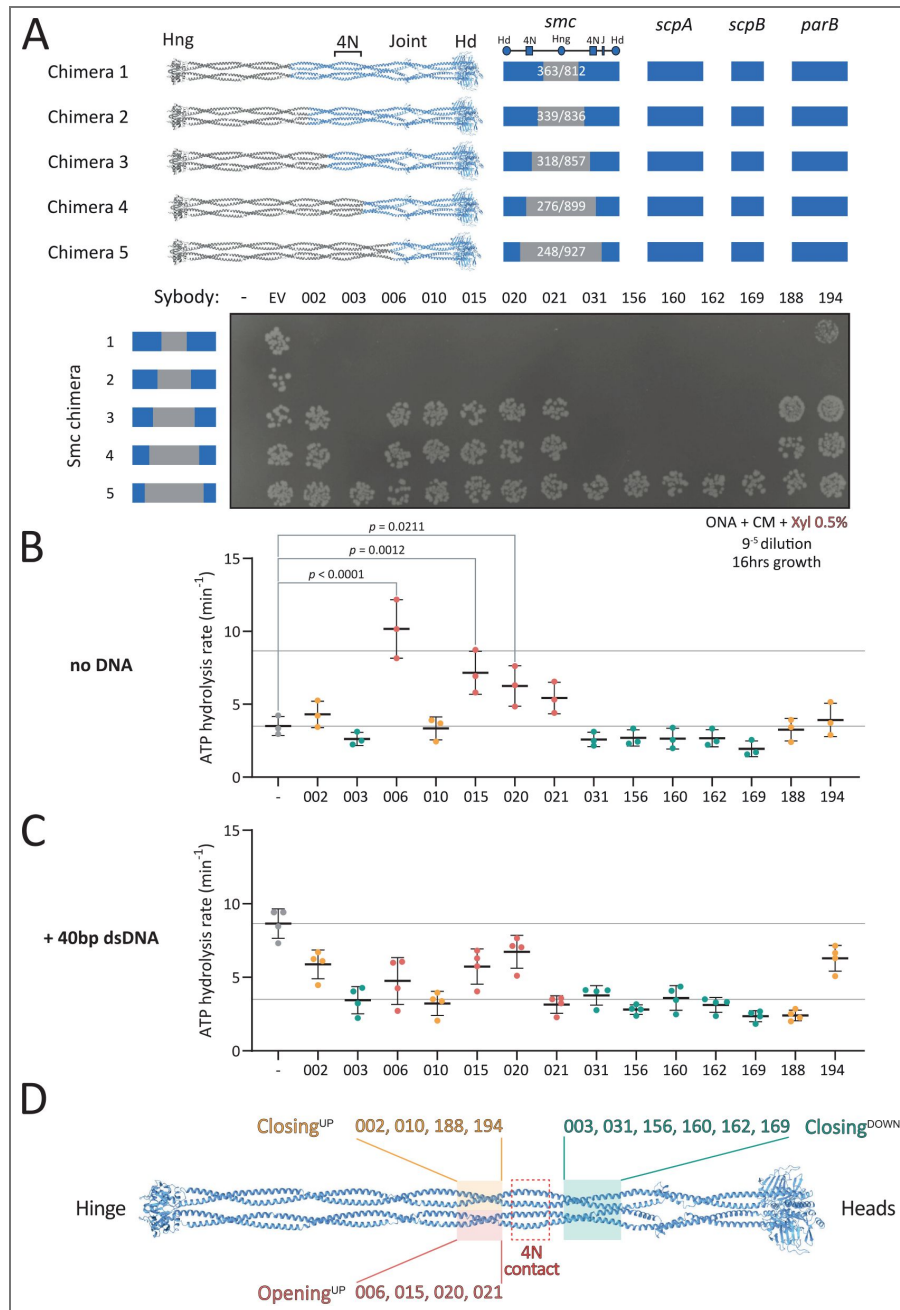


Figure 3. Mapping sybody binding sites on Smc-ScpAB.

(A) Colony formation of *B. subtilis* strains expressing chimeric Smc-ScpAB complexes comprising *S. pneumoniae* and *B. subtilis* sequences. The schematic above the spot assay depicts the species origin of the *smc* gene, the *scpAB* operon and *parB* gene (blue: *B. subtilis*; grey: *S. pneumoniae*). “-” indicates no insertion at the *amyE* locus; “EV” refers to the empty vector control containing only the chloramphenicol resistance cassette that was inserted in *amyE*; numbered labels correspond to sybodies. Cells were spotted on rich medium (ONA) supplemented with chloramphenicol and xylose and incubated for 16 h at 37 °C. Hd: head, Hng: hinge, 4N: 4N arm-to-arm contact, J: joint. **(B)** ATP hydrolysis rates of *bsuSmc-ScpAB* in the presence of sybodies but absence of DNA. Significant effects by one-way ANOVA are indicated by p values. **(C)** ATPase rates in the presence of 40 bp dsDNA. All sybodies significantly reduced DNA-stimulated ATP hydrolysis. Reported p-values: Sb020 ($p = 0.340$), Sb194 ($p = 0.0049$), Sb002 ($p = 0.0007$), Sb015 ($p = 0.0003$); all others, $p < 0.0001$. **(D)** Schematic summary of sybody binding sites mapped onto the Smc dimer, categorized by their effect on ATPase activity. Sybodies are grouped based on functional impact and mapped to corresponding structural regions: pink/yellow boxes indicate residues 318–339 and 836–857; green boxes mark residues 248–276 and 899–927; and the red box highlights the 4N contact region (approx. residues 290–320).

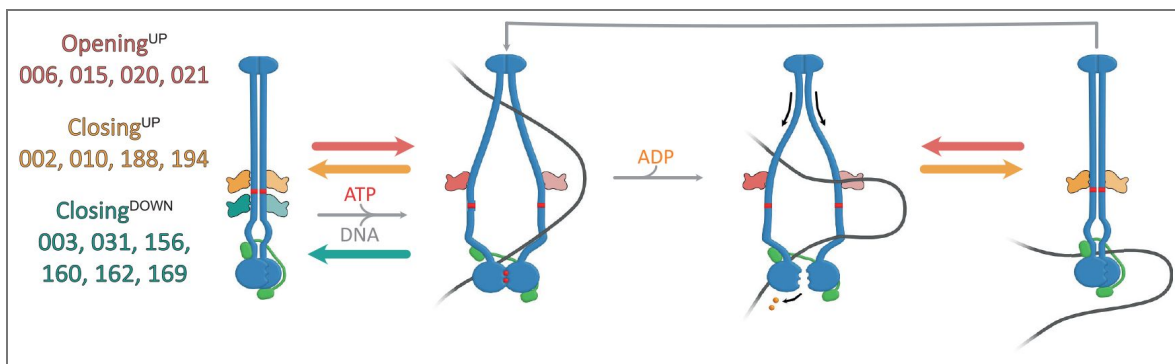


Figure 4. Proposed models for sybody interactions with *bsuSmc-ScpAB* during the ATPase cycle.

Model for sybodies of the Opening^{UP}, Closing^{UP} and Closing^{DOWN} group. Opening^{UP} sybodies, likely prevent complete arm closure. In the presence of DNA, these sybodies may stabilize a suboptimal conformation for head disengagement, resulting in reduced ATPase activity (not shown here). Closing^{UP} and Closing^{DOWN} sybodies stabilize a closed-arm conformation. These likely hinder DNA segments from entering the inter-arm space and accessing the hinge-proximal DNA binding site.

engagement. Curiously, addition of DNA in the presence of Sb006 and Sb021 reduced—rather than stimulated—ATPase activity (Figure 3B–C), suggesting that concurrent DNA and sybody binding may trap the complex in a non-productive conformation, perhaps by favoring the open state excessively. These findings are consistent with the idea that 4N contact region plays a critical role in arm opening and ATPase stimulation.

Interestingly, the observed ATPase profiles correlate with distinct binding regions along the Smc arms. All ATPase-stimulating sybodies (Sb006, Sb015, Sb020, Sb021) bind above the 4N contact (“Opening^{UP}”), while sybodies that hinder DNA stimulation fall into two subgroups: those binding above the 4N contact (Sb002, Sb010, Sb188, Sb194—“Closing^{UP}”) and those binding below it (Sb003, Sb031, Sb156, Sb160, Sb162, Sb169—“Closing^{DOWN}”) (Figure 3D). These results indicate that the sybodies perturb *bsu*Smc-ScpAB in two principal ways: either by favoring the opening of the arms or by constraining the arms in a more closed state. The results highlight arm opening and closure as central regulatory features of bacterial Smc-ScpAB (Soh et al., 2015; Minnen et al., 2016; Diebold-Durand et al., 2017; Vazquez Nunez, Ruiz Avila and Gruber, 2019) and confirm the key role of the 4N arm contact in balancing the opening and closing of SMC arms (Vazquez Nunez et al., 2021).

Discussion

Over the past three decades, sybodies and their synthetic counterparts have transformed biomedical research and structural biology by enabling the targeting and stabilization of transient protein conformations. Despite these advances, their application to probing protein function in living cells has remained limited (Seeger et al., 2012; Deneka et al., 2021). Here, we expand the utility of synthetic sybody selection by employing it as a tool to screen for genetic probes targeting the function of selected molecular machines *in vivo*. This strategy combines the advantages of controlled biochemical reconstitution with the ability to study the function of proteins in their native context. Importantly, sybody selection requires no prior knowledge of vulnerable regions, providing an unbiased means to interrogate protein function that is difficult to achieve through rational design. Our study based on the *B. subtilis* Smc-ScpAB DNA motor as a model uncovers a central regulatory region in the coiled coil arms of Smc and highlight how distinct arm conformations shape SMC function in loop extrusion and chromosome segregation.

Revealing Smc arm dynamics through synthetic binders

All fourteen Smc-disrupting sybodies share the loop library design. Moreover, their Smc binding sites map to the two short coiled coil segments flanking the conserved 4N arm-to-arm contact, a region previously implicated in key conformational transitions during arm opening (Vazquez Nunez et al., 2021). While the fourteen sybodies harbour different epitope binding sequences, we suspect that they may all bind to the Smc dimer in an analogous manner. With the long CDR3 loops known to bind narrow structural cavities (Rasmussen et al., 2011; Kruse et al., 2013), we speculate that these sybodies may intercalate between the two coiled coil arms (Figure 3D). Structural studies are needed to test this hypothesis directly, however, our attempts at X-ray crystallography and electron microscopy of Smc-sybody complexes have failed so far.

Notably, while the coiled coil dimer appears similarly structured from the elbow to the hinge (Diebold-Durand 2017) (Figure 3D), no Smc-disrupting sybodies mapped to regions further away from the 4N contact (Figure 3D). This indicates that opening and closure of the Smc arms might be dispensable at these places, although formal testing would require isolating binders efficiently targeting these regions, potentially facilitated by binder design, and expressing them in *B. subtilis*. This parallels with findings with the bacterial SMC defense system Wadjet, where hinge-proximal arm opening is dispensable for loop extrusion, but essential for obstacle bypass (Liu et al., 2025). Alternatively, these arm-to-arm contacts might be more stable and thus less likely to be interfered with by sybody binding. Similarly, sybodies targeting below the 4N contact all reduce the ATPase rate (or rather its DNA stimulation), indicating that artificial opening at this position

may not disrupt Smc function (Figure S5 [↗](#)). Again, these hypotheses need to be tested more directly. We envision that the design of protein binders may allow us to confirm these observations and further dissect the underlying mechanisms.

This work establishes sybodies as precision tools for blocking ATP-driven machines inside bacterial cells. By targeting allosteric control points in Smc, they enable mechanistic dissection of loop extrusion and open new avenues for studying dynamic complexes *in vivo*. More broadly, the study demonstrates how synthetic binders can trap, stabilize, or block transient conformations of active chromatin-associated machines, providing a powerful means to probe their mechanisms in living cells. Looking ahead, the rational design of protein binders with tailored geometry and allosteric potential could allow researchers to manipulate and visualize specific conformational states with even greater control and refinement, across both prokaryotic and eukaryotic organisms, while the generation and *in vivo* screening of sybodies remains an attractive approach to gain unexpected biological insights in an unbiased manner.

Material and methods

Bacillus subtilis strains construction and growth

B. subtilis strains used in this study were derived from the parental strain 1A700. Allelic replacement was achieved via double-crossover recombination at the endogenous *smc* locus using the natural competence method described by (Bürmann et al., 2013 [↗](#)). Chromosome-integration of sybody genes were performed by transforming a vector containing an *amyE* homologous regions on each side of the recombinant gene of interest, as described in (Diebold-Durand, Bürmann and Gruber, 2019 [↗](#)). Transformants were selected on ONA solid medium containing the appropriate antibiotics. Following transformation, strains were purified by single colony isolation and verified through a combination of marker testing, phenotype assessment, PCR, and Sanger sequencing of the *smc* locus, when appropriate. For spot assays, cells were cultured to stationary phase in liquid LB medium, and 9^{-2} and 9^{-5} dilutions were then spotted onto solid ONA medium with chloramphenicol (5 ug/mL) and xylose (0.5% w/v) when induction of the P_{xyl} promoter was required. The strains, primers and plasmids used in this study are listed in Table S3, Table S4 and Table S5 [↗](#), respectively.

For viability assessment, 200 LB medium was inoculated with a 1:1000 dilution of a dense *B. subtilis* culture and grown to stationary phase. The culture was then serially diluted in LB medium. From each dilution (9^{-1} to 9^{-8}), 5 μ L was spotted onto LB agar plates containing the appropriate antibiotic selection. Plates were incubated, and viability was assessed by imaging the spots at 16- and 24-hours post-incubation.

Sybody selection

Sybody selection was performed following the protocol by (Zimmermann et al., 2020 [↗](#)), with modifications to enrich binders specific for the ATP/DNA-stabilized “open” form of the *bsu*Smc-ScpAB complex.

Ribosome Display

The standard WTB buffer (50 mM Tris/acetate pH7.4, 150 mM NaCl, 50 mM MgAc₂) and its derivatives were adjusted to 50 mM NaCl to facilitate *bsu*Smc-ScpAB binding to dsDNA. The target complex was assembled by mixing 500 nM *bsu*Smc(C119S, C437S, C826S, E1118Q, R643C)-ScpAB, 2 mM ATP, and 5 μ M 40 bp dsDNA in WTB-D-BSA buffer (WTB including 0.5% (w/v) BSA and 0.1% Tween-20), incubated for 15 min at room temperature (RT). The ribosome display panning solution was supplemented with 2 mM ATP and 5 μ M dsDNA to maintain binding conditions. Solution panning and biotinylated target capture were conducted at RT to optimize selection for the “open” complex. Washing steps incorporated 2 mM ATP and 5 μ M 40 bp dsDNA, while elution, reverse transcription, and cDNA amplification followed the original protocol. Reverse transcribed RNA molecules were quantified by qPCR.

First round of phage display

Phagemid libraries were cloned and electroporated into *E. coli* SS320 as previously described (Zimmermann et al., 2020 [DOI](#)). Phage production was performed using M13KO7 helper phage. The phage display buffer (TBSM) consisted of 20 mM Tris-HCl (pH 7.4), 50 mM NaCl, and 2 mM MgCl₂. Target preparation involved incubating 500 nM biotinylated *bsuSmc*(C119S, C437S, C826S, E1118Q, R643C)ScpAB with 2 mM ATP and 5 μM dsDNA in TBSM-BSA-D (TBSM including 0.5% (w/v) BSA and 0.05% Tween-20). Phages (10¹²/ml) were incubated with the target at 50 nM for 20 min at RT before immobilization on neutravidin-coated plates. Washing (with added 2 mM ATP and 5 μM 40 bp dsDNA) and elution otherwise followed the original protocol, with enrichment assessed by qPCR. Amplified phages were used for a second selection round.

Second round of phage display

Purified phages from the first round were used at 5 × 10¹³ phages/ml, and the target was prepared as in the first round. Phage-target binding occurred at 50 nM in the presence of ATP and dsDNA, followed by capture on magnetic beads. A competition step using non-biotinylated *bsuSmc*-ScpAB (1 μM) was included before washes with TBSM-D. Phages were eluted and enrichment was determined by qPCR. Phagemids were purified, sub-cloned into the pNb_init vector, and transformed into *E. coli* MC1061.

Growth curve analyses

Growth curves were generated from *B. subtilis* strains grown overnight to exponential phase in LB medium supplemented with 0.5% glucose, incubated at 30°C with shaking. The following day, fresh 5 mL LB cultures were inoculated with the overnight culture to an OD of 0.005 and incubated at 37°C with shaking until reaching an optical density (OD) of 0.05. Subsequently, each culture was subjected to a twofold dilution in a 96-well plate (Costar #3596), achieving a final volume of 200 μL per well. Where necessary, cultures were induced with 0.5% xylose. Plates were incubated at 37°C with continuous shaking in a Thermo Scientific Multiskan FC plate reader. Growth curves were determined by light scattering at 620 nm. The BactEXTRACT app was used to perform Analysis and visualization of the data (Dénéréaz and Veening, 2024 [DOI](#)).

Protein purification

Purification of *bsuSmc* and biotinylated *bsuSmc*(C119S, C437S, C826S, E1118Q, R643C)

bsuSmc proteins were purified according to (Bürmann et al., 2017 [DOI](#)). pET-22 or pET-28 plasmids encoding the *Smc* recombinant sequences were transformed into *E. coli* BL21-Gold (DE3) cells. Protein expression was carried out in ZYM-5052 autoinduction medium for 23 hours at 24°C. The cells were harvested and resuspended in lysis buffer (50 mM Tris-HCl, pH 7.5, 150 mM NaCl, 1 mM EDTA, 1 mM DTT, 10% (w/v) sucrose) supplemented with protease inhibitor cocktail. Cell lysis was achieved by sonication, and the lysate was clarified by centrifugation. The supernatant was filtered through a 0.45 μm membrane and loaded onto two HiTrap Blue HP 5 mL columns connected in series. Elution was performed using lysis buffer with 1 M NaCl. The main peak fractions were pooled and diluted with salt-free buffer (50 mM Tris-HCl, pH 7.5; 1 mM EDTA; 1 mM DTT) to a final conductivity equivalent to 50 mM NaCl (~8 mS/cm). This diluted sample, supplemented with protease inhibitor cocktail, was applied to a HiTrap Heparin HP 5 mL column, and the proteins were eluted using a linear gradient up to 2 M NaCl. The main peak fractions (~ 5 mL) were collected and subjected to further purification by gel filtration using an XK 16/70 Superose 6 PG column equilibrated with 50 mM Tris-HCl, pH 7.5, 200 mM NaCl, 1 mM EDTA, and 1 mM TCEP. The peak fractions were collected, concentrated using a Vivaspin 15 10K MWCO filter, flash frozen in liquid nitrogen, and stored at -80°C. Protein concentration was determined by absorbance, utilizing theoretical molar absorption and molecular weight values.

For the biotinylated *bsuSmc*(C119S, C437S, C826S, E1118Q, R643C) protein used in the sybody selection, the protein was first purified as described above, except that the reducing agent (TCEP) was omitted from the final gel filtration buffer. Labelling was performed by incubating 600 μL of

bsuSmc(C119S, C437S, C826S, E1118Q, R643C) (83 μ M in 50 mM Tris-HCl pH 7.5, 200 mM NaCl) with 1 mM PEG2-biotin maleimide for 10 minutes at 4 °C. The reaction was quenched by addition of 0.5 mM 2-mercaptoethanol. Excess label was removed using Zeba spin desalting columns (Thermo Fisher) in multiple parallel runs due to volume limitations. The final yield was 600 μ L at 72.35 μ M (subunit concentration). Labelled *bsuSmc*(C119S, C437S, C826S, E1118Q, R643C) was loaded onto a Superose 6 10/300 Increase column pre-equilibrated in 50 mM Tris-HCl pH 7.5, 200 mM NaCl. Peak fractions were pooled and mixed with ScpA and ScpB at a 1:1:2 molar ratio relative to the *bsuSmc* dimer. The final concentration of the full complex was adjusted to 5.7 μ M. Complexes were aliquoted in 100 μ L portions, flash-frozen in liquid nitrogen, and stored at -80 °C.

Purification of *bsuScpA*

ScpA was purified using the method described by (Vazquez Nunez, Ruiz Avila and Gruber, 2019 [↗](#)). *E. coli* BL21-Gold (DE3) cells, transformed with a pET-28 derived plasmid encoding the ScpA protein, were used for expression. Cultivation was performed in ZYM-5052 autoinduction medium at 16°C for 28 hours. Cells were then harvested and resuspended in lysis buffer (50 mM Tris-HCl, pH 7.5; 200 mM NaCl; 5% glycerol) supplemented with protease inhibitor cocktail. The cells were lysed by sonication, and the lysate was clarified by centrifugation. The supernatant was applied to a 5 mL HiTrap Q ion exchange column and eluted with a gradient up to 2 M NaCl. Peak fractions were pooled and adjusted to a final concentration of 3 M NaCl by mixing with 4 M NaCl buffer. This mixture was loaded onto a HiTrap Butyl HP column and eluted with a reverse gradient to 50 mM NaCl. Eluted peak fractions were concentrated to 5 mL using Vivaspin 15 10K MWCO filters and further purified by size exclusion chromatography (SEC) on a HiLoad 16/600 Superdex 75 pg column equilibrated with 20 mM Tris-HCl, pH 7.5, and 200 mM NaCl. The purified protein was concentrated, flash frozen, and stored at -80°C.

Purification of *bsuScpB*

ScpB was purified following the protocol outlined by (Vazquez Nunez, Ruiz Avila and Gruber, 2019 [↗](#)). The coding sequence of ScpB, cloned into a pET-22 derived plasmid, was transformed into chemically competent BL21-Gold (DE3) *E. coli* cells. These cells were cultivated in ZYM-5052 autoinduction medium at 24°C for 23 hours. Subsequently, the cells were harvested and resuspended in lysis buffer (50 mM Tris-HCl, pH 7.5; 150 mM NaCl; 1 mM EDTA; 1 mM DTT) supplemented with protease inhibitor cocktail. Cell lysis was performed by sonication, followed by centrifugation to remove cell debris. The resulting supernatant was diluted to a final NaCl concentration of 50 mM and loaded onto a 5 mL HiTrap Q HP column. Elution was achieved using a gradient up to 2 M NaCl. The eluate was then diluted with lysis buffer containing 4 M NaCl to achieve a final concentration of 3 M NaCl. This sample was applied to two 5 mL HiTrap Butyl columns connected in series, and the protein was eluted with a reverse gradient to 50 mM NaCl. The peak fractions from this column were concentrated and subjected to size exclusion chromatography (SEC) using a HiLoad 16/600 Superdex 200 pg column equilibrated with 50 mM Tris-HCl, pH 7.5, 100 mM NaCl, and 1 mM DTT. Fractions containing ScpB were concentrated, flash frozen, and stored at -80°C.

Medium scale sybody purification

Individual sybody plasmids were transformed into chemically competent *E. coli* MC1061 cells via heat shock at 42 °C for 45 s, followed by recovery in LB medium at 37 °C. Transformed cells were cultured overnight in TB medium supplemented with 25 μ g/ml chloramphenicol. Precultures were used to inoculate 50 mL TB cultures, which were grown at 37 °C before shifting to 22 °C. Expression was induced with 0.02% (wt/vol) L-(+)-arabinose and continued overnight at 22 °C with shaking. Cells were harvested by centrifugation and resuspended in periplasmic extraction buffer. Following incubation at 4 °C, cells were pelleted, and the supernatant was supplemented with imidazole to 15 mM. The extract was incubated with His MultiTrap HP resin, followed by centrifugation and washing with TBS containing 30 mM imidazole. Elution was performed using TBS with 300 mM imidazole. Purified sybodies were analyzed on a Sepax SRT-10C SEC100 column at 1 ml/min flow rate. Monomeric sybodies eluted between 11–12.5 ml, while retention volumes

<11 ml indicated oligomerization, and >14 ml suggested strong column interaction. Non-expressed, oligomeric, or highly interacting sybodies were discarded. Typical yields ranged from 200 µg to 1 mg (Zimmermann et al., 2020 [DOI](#)).

Fluorescence imaging of ParB-GFP strains Image acquisition

B. subtilis cells were first cultured in LB supplemented with 5 µg/mL chloramphenicol and 0.5% glucose at 30°C, from which a day culture was inoculated to 0.005 in LB with 5 µg/mL chloramphenicol. When needed, cultures were induced at OD₆₀₀ ~ 0.02 with 0.5% xylose and grown an extra 30 minutes until imaging at an OD₆₀₀ of 0.04. For microscopy analysis, 0.5 µL of the cell suspension was spotted onto agarose-coated microscopy slides. Images were acquired using a Leica DMI8 microscope equipped with an sCMOS DFC9000 (Leica) camera, a SOLA light engine (Lumencor), and a 100×/1.40 oil-immersion objective. Exposure time for image acquisition was set to 600 ms. Acquired images were processed using LAS X Office software (v.1.4.7.28982, Leica Microsystems).

Image analysis

Microscopy images were analyzed using a fully automated image processing pipeline. Cell segmentation was performed with the pretrained Omnipose model (Cutler et al., 2022 [DOI](#)) implemented in Cellpose (Stringer et al., 2021 [DOI](#)), which is optimized for bacterial morphologies. Prior to segmentation, a normalization step was applied to standardize image contrast. Following segmentation, intracellular foci were detected using Spotiflow (Mantes et al., 2024 [DOI](#)), a dedicated model trained for spot identification. Each detected spot was assigned to an individual cell based on spatial location, enabling quantification of the number of foci per cell. Cell morphology analysis was also performed to estimate cell length.

Quantitative results were compiled into a structured dataset, including the number of cells per image, percentage of cells lacking foci, number of foci per cell, cell length, and number of foci per micron. Data visualization and statistical analysis was performed using GraphPad Prism 10.4.1 (627).

ATPase measurements

ATPase activity was assessed using the coupled pyruvate kinase/lactate dehydrogenase reaction as described by (Bürmann et al., 2017 [DOI](#)). ADP production was monitored over a period of 1 hour by measuring the absorbance changes of NADH at 340 nm. Data collection was performed using a Synergy Neo Hybrid Multi-Mode Microplate Reader. The reaction mixture comprised 1 mM NADH, 3 mM phosphoenolpyruvic acid, 100 U pyruvate kinase, 20 U lactate dehydrogenase, and varying concentrations of ATP. For assays requiring double-stranded oligonucleotides, a 40 bp oligonucleotide (5'-TTAGTTGTTC GTAGTGCTCG TCTGGCTCTG GATTACCCGC-3') was added to a final concentration of 3 mM. The final protein concentration in the assay was 0.15 µM *bsu*Smc dimers in ATPase assay buffer (50 mM HEPES-KOH, pH 7.5; 50 mM NaCl; 2 mM MgCl₂). All measurements were conducted at 25°C.

Data availability

All raw data will be made available at Mendeley Data DOI: [10.17632/k6k62p7z2s.1](https://doi.org/10.17632/k6k62p7z2s.1) [DOI](#).

Supplementary figures

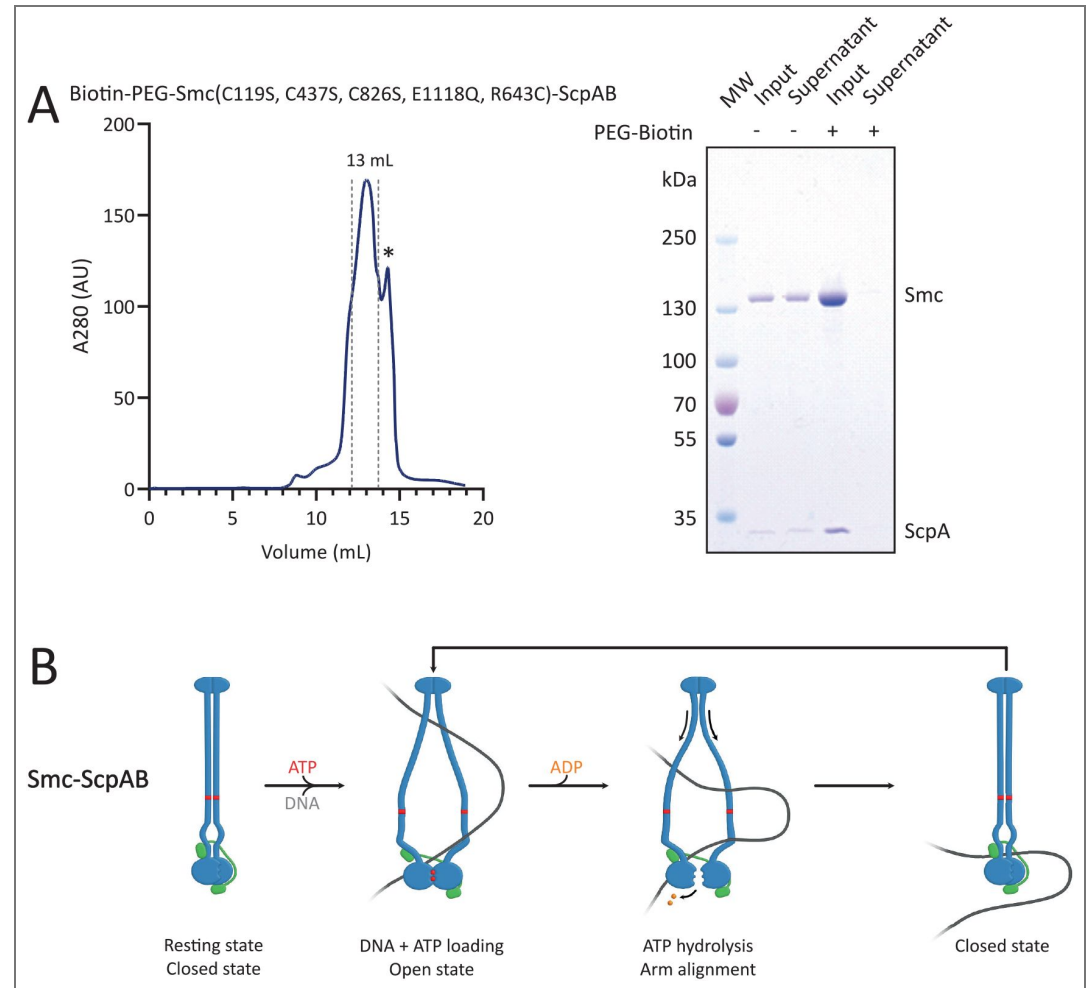


Figure S1. Preparation of *bsuSmc*(C119S, C437S, C826S, E1118Q, R643C)-ScpAB complex and loop extrusion model. (A) Size-exclusion chromatography profile of biotinylated *bsuSmc*(C119S, C437S, C826S, E1118Q, R643C)-ScpAB complex. Peak fractions (elution at ~13 mL, pooled fractions indicated by dotted lines for a final volume of 1.8 mL) were collected for downstream use; the final concentration was 5.9 μ M (dimer). * : A secondary peak appeared at ~14 mL; its identity was uncertain and was excluded from the peak. SDS-PAGE confirms successful biotinylation of the *bsuSmc*(C119S, C437S, C826S, E1118Q, R643C)-ScpAB complex. **(B)** Segment-capture model for *bsuSmc*-ScpAB, in which ATP binding and hydrolysis drive transitions between open and closed conformations to mediate loop extrusion.

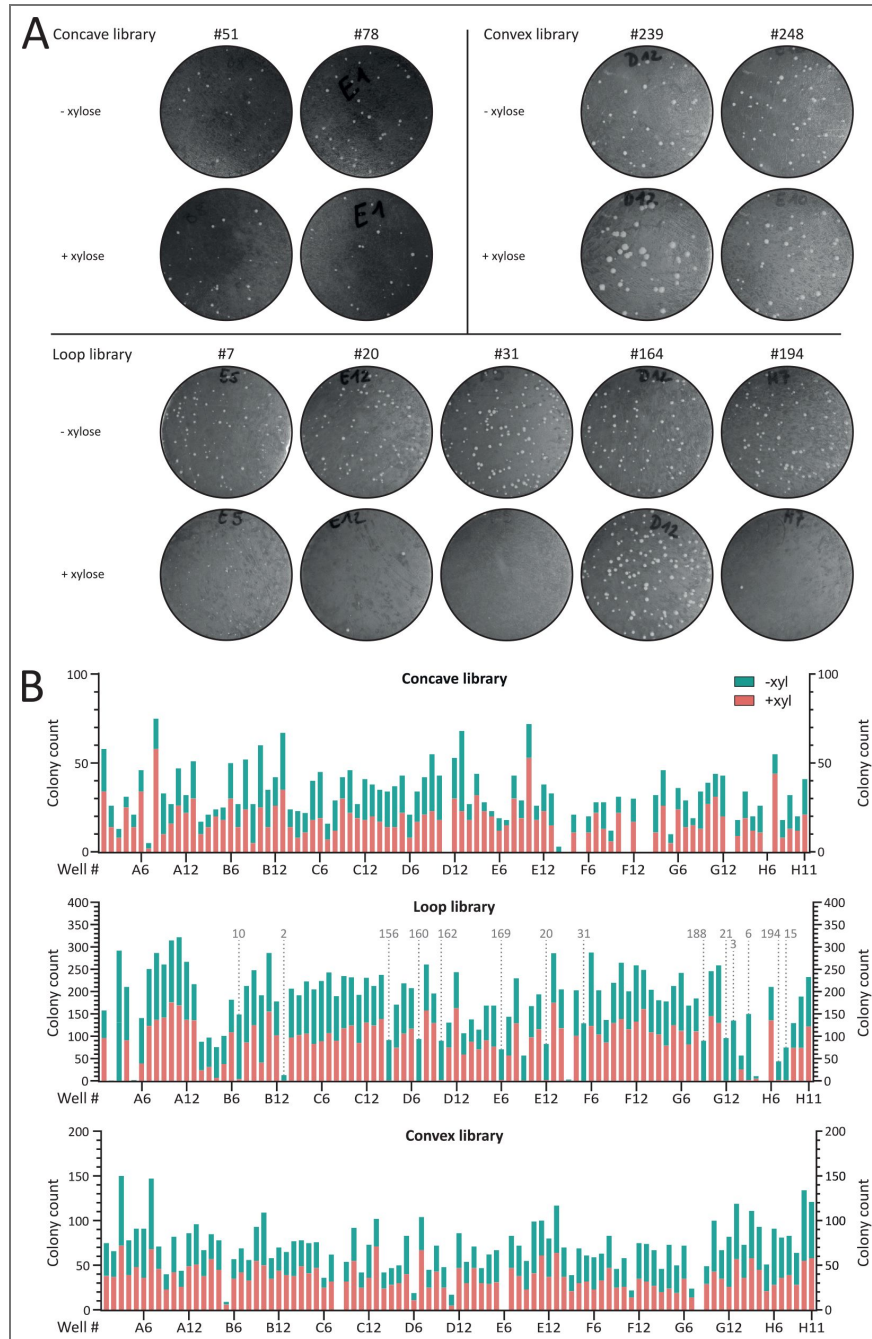


Figure S2. Representative results from *in vivo* sybody selection based on colony formation.

(A) Representative examples of *in vivo* transformation assays for nine sybody-expressing *B. subtilis* strains. Sybody genes were integrated into the *amyE* locus under xylose-inducible control. Transformants were grown on oxid nutrient agar (ONA) plates supplemented with chloramphenicol and 0.5% xylose to assess growth defects. Shown are candidates from the concave library: Sb051 (well B08) and Sb078 (D12). Loop library: Sb007 (E05), Sb020 (E12), Sb031 (F05), Sb164 (D12) and NbH07 (Sb194). Convex library: Sb239 (D12) and Sb248 (E10). **(B)** Growth assay results for the 285 *B. subtilis* strains expressing individual xylose-inducible sybodies. Bars show each colony counts “without” (green) on top of “with” (pink) xylose. Strains are ordered by their original 96-well plate. Only the loop library yielded sybodies that consistently impaired growth under inducing conditions (dotted line). Sybody numbers indicated above the plots correspond to positive candidates used in subsequent experiments, numbered according to order of use. Notably, the strain corresponding to the E09 M sybody (Sb018), also showed an absence of growth upon sybody induction. However, this sybody candidate gave intermediate phenotypes in later experiments, which is why it was excluded from this study. Results for the loop library are also shown in Figure 1.

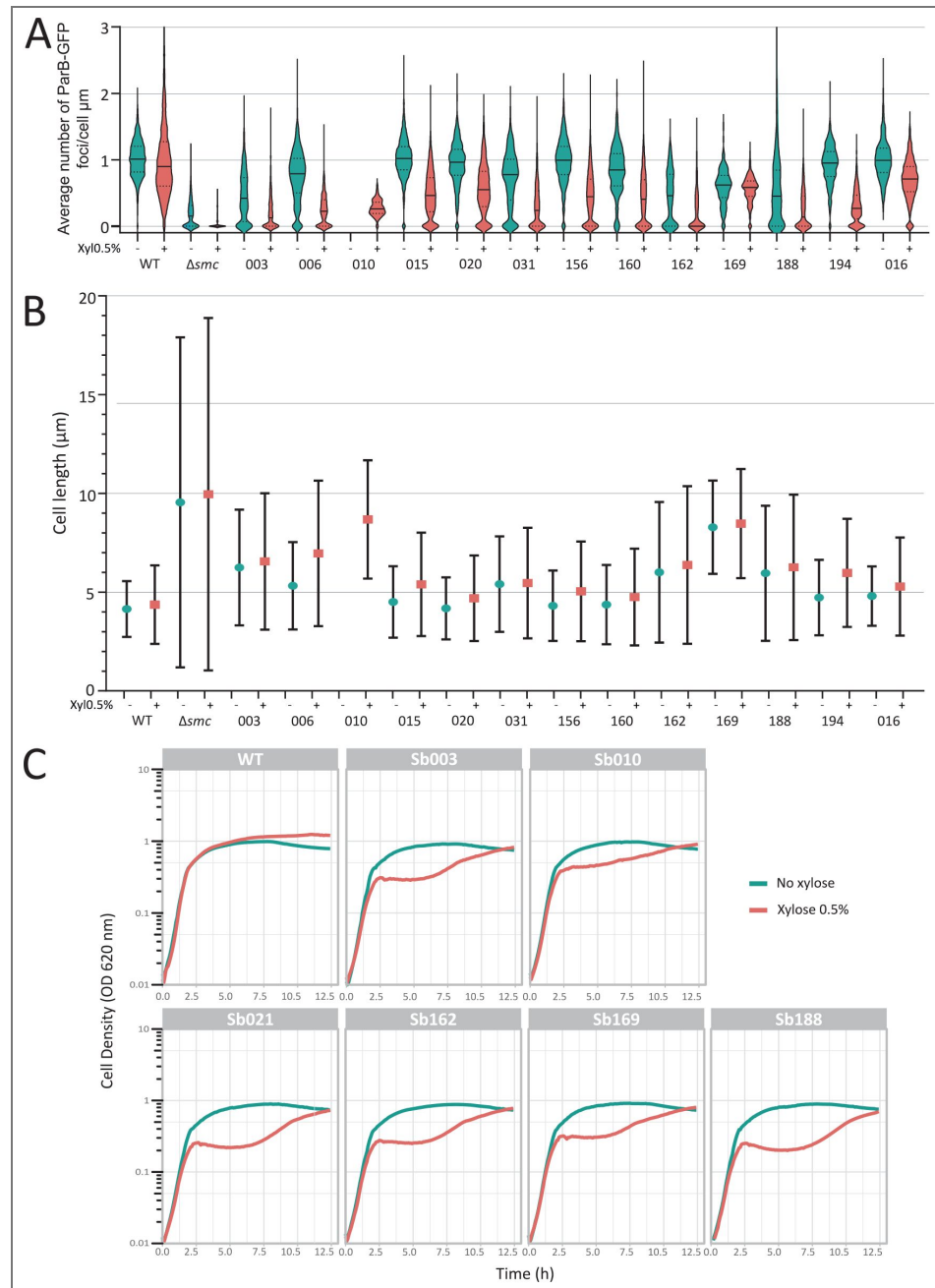


Figure S3. Functional impact of sybody expression on chromosome organization, cell length, and growth in *B. subtilis*.

(A) Average number of ParB-GFP foci per μm of cell length in strains expressing individual sybodies, with and without xylose induction. Most uninduced strains differed significantly from the SMC WT control, likely due to leaky expression from the P_{xy} ($p < 0.0001$), except Sb015 ($p > 0.9999$), Sb156 ($p = 0.2345$), and Sb016 (negative control, $p > 0.9999$). **(B)** Average cell length across the same conditions, measured in ParB-GFP strains with or without SMC, and with or without each sybody. Bars indicate standard deviation. Tested sybodies caused cell elongation, a characteristic phenotype of impaired SMC activity, as chromosome segregation defects delay cell division, with mean lengths from $4.70 \pm 2.16 \mu m$ (Sb020) to $8.68 \pm 2.99 \mu m$ (Sb010), compared to $4.51 \pm 1.95 \mu m$ in wild-type and $9.96 \pm 8.92 \mu m$ in Δsmc . **(C)** Growth curves of *B. subtilis* strains expressing individual sybodies compared to the WT strain, with and without xylose induction. Each curve represents the mean of two biological replicates. Six sybodies were randomly picked and showed in this figure. Fluorophore-free strains showed wild type-like growth without xylose but strong delays post-induction, followed by partial growth recovery after 12 hours, likely due to xylose depletion or suppressor emergence. In sybody-expressing strains, a clear drop in cell density was observed ~ 2.5 hours post-induction.

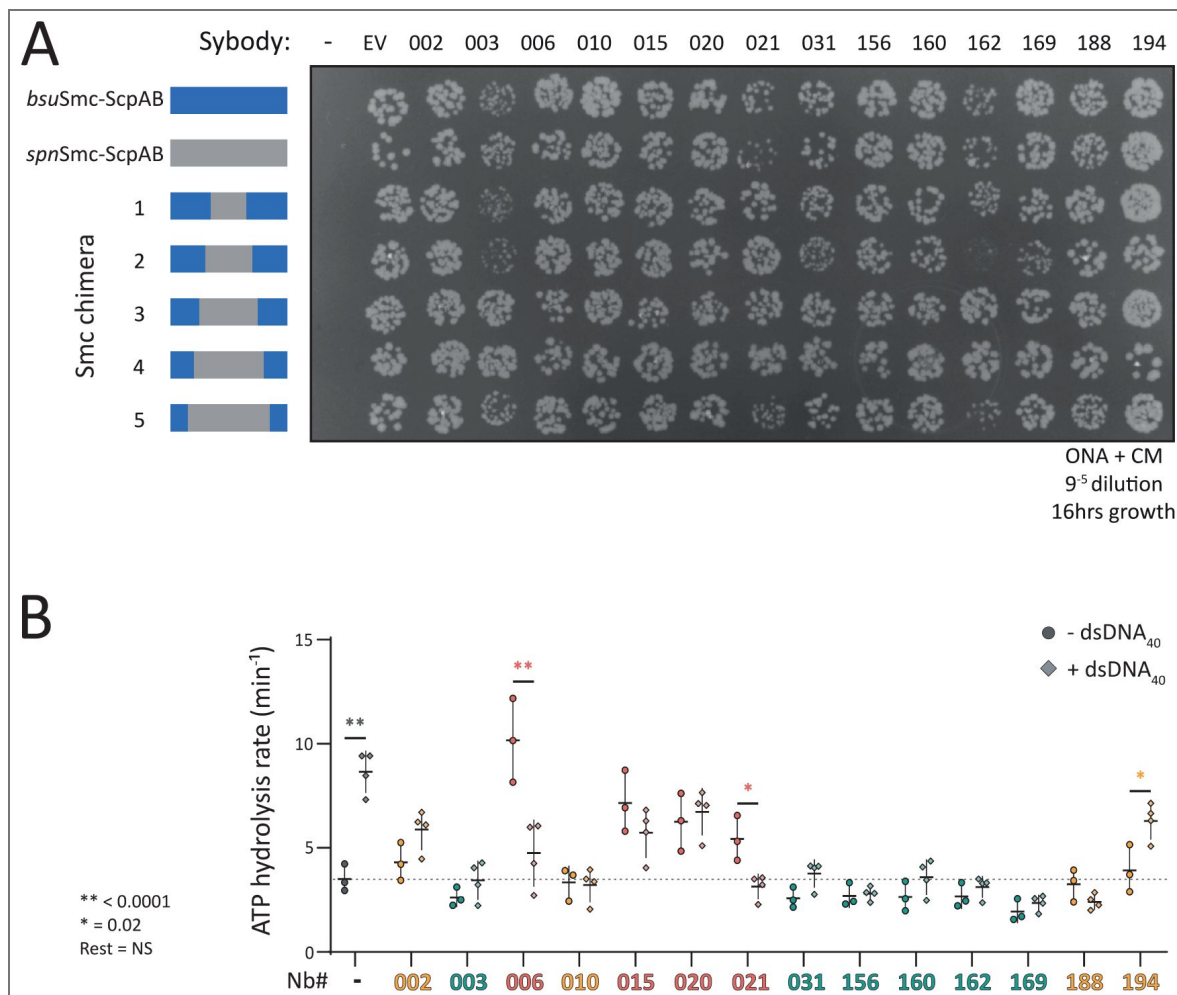


Figure S4. ATPase hydrolysis rate of the *bsuSmc-ScpAB* complex with each selected sybody and viability of *B. subtilis* strains expressing chimeric Smc-ScpAB complexes in the absence of sybody induction.

(A) Spotting assay assessing the growth of *B. subtilis* strains carrying chimeric Smc-ScpAB operons composed of *S. pneumoniae* and *B. subtilis* components. Strains harbor individual sybody constructs integrated at the *amyE* locus under control of a xylose-inducible promoter but were grown without inducer. “-” indicates no integration at *amyE*; “EV” corresponds to an empty vector control containing only the chloramphenicol resistance cassette. Numbered labels indicate strains carrying specific sybody constructs. Cells were spotted on rich medium (ONA) containing chloramphenicol and incubated for 16 h at 37 °C. (B) Overall ATPase hydrolysis rate of the *bsuSmc-ScpAB* complex in presence of each sybody and +/-DNA. Sb06, 021 and 194 triggered a significant difference in ATP hydrolysis between +/-dsDNA conditions. The rest is non-significant.

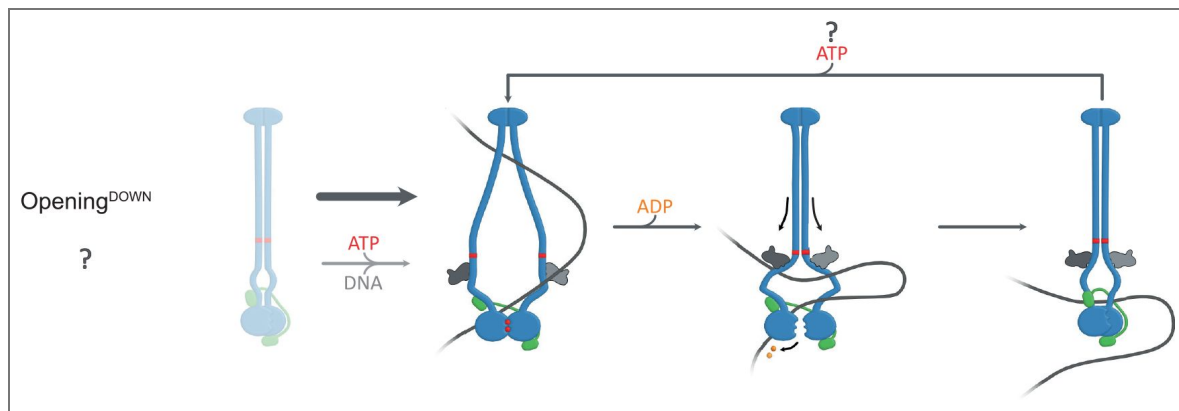


Figure S5. Hypothetical model for an $\text{Opening}^{\text{DOWN}}$ class of sybodies, not recovered in this study.

This may indicate that stabilizing an open conformation near the ATPase heads does not interfere with Smc function.

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Additional information

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Author contributions

O.G. performed all *in vivo* experiments from sybody selection, fluorescence microscopy imaging, chimeric phenotypic screen. M.T. and L.H.-H. performed the *in vitro* sybody screening (ribosome display, phage display, and first sybody purification). O.G. performed protein purifications, cross-linking and ATPase measurements, as well as data analysis, figures and writing. O.G. wrote the first manuscript draft. All authors reviewed and edited the manuscript. Funding acquisition and supervision, S.G.

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Additional files

Table 1. [↗](#) Sybody enrichments at different steps of the selection procedure. Ribosome display output was quantified by qPCR, while phage display results include final phage titers and enrichment values from rounds one and two, also measured by qPCR.

Table S1. [↗](#) Sybody sequences.

Table S2. [↗](#) Percentage of cells lacking ParB-GFP foci, indicating the absence of chromosome due to segregation defect. In presence and absence of xylose for sybody induction.

Table S3, S4 and S5. [↗](#) Table S3: Strains. Table S4: Plasmids. Table S5: Oligonucleotides

References

Bock F.P., et al. (2022) A joint-ParB interface promotes Smc DNA recruitment. *Cell reports* **40**:111273 <https://doi.org/10.1016/j.CELREP.2022.111273> | PubMed

Bürmann F., et al. (2013) An asymmetric SMC-kleisin bridge in prokaryotic condensin. *Nature Structural & Molecular Biology* **20**:371-379 <https://doi.org/10.1038/nsmb.2488> | PubMed

Bürmann F., et al. (2017) Tuned SMC Arms Drive Chromosomal Loading of Prokaryotic Condensin. *Molecular Cell* **65**:861-872.e9. <https://doi.org/10.1016/j.molcel.2017.01.026> | PubMed

- Bürmann F., Gruber S. (2015) SMC condensin: promoting cohesion of replicon arms. *Nature Structural & Molecular Biology* 22:9 22:653-655 <https://doi.org/10.1038/nsmb.3082> | PubMed
- Cutler K.J., et al. (2022) Omnipose: a high-precision morphology-independent solution for bacterial cell segmentation. *Nature Methods* 19:1438-1448 <https://doi.org/10.1038/s41592-022-01639-4> | PubMed
- Deneka D., et al. (2021) Allosteric modulation of LRRC8 channels by targeting their cytoplasmic domains. *Nature Communications* 12:5435 <https://doi.org/10.1038/s41467-021-25742-w> | PubMed
- Dénéréaz J., Veening J.-W. (2024) BactEXTRACT: an R Shiny app to quickly extract, plot and analyse bacterial growth and gene expression data. *Access Microbiology* 6:000742.v3 <https://doi.org/10.1099/acmi.0.000742.v3> | PubMed
- Diebold-Durand M.L., et al. (2017) Structure of Full-Length SMC and Rearrangements Required for Chromosome Organization. *Molecular Cell* 67:334-347.e5. <https://doi.org/10.1016/j.molcel.2017.06.010> | PubMed
- Diebold-Durand M.-L., Bürmann F., Gruber S. (2019) High-Throughput Allelic Replacement Screening in *Bacillus subtilis*. *Methods in Molecular Biology* 2004:49-61 https://doi.org/10.1007/978-1-4939-9520-2_5 | PubMed
- Gabler F., et al. (2020) Protein Sequence Analysis Using the MPI Bioinformatics Toolkit. *Current Protocols in Bioinformatics* 72:e108 <https://doi.org/10.1002/cpbi.108> | PubMed
- Gruber S., et al. (2014) Interlinked Sister Chromosomes Arise in the Absence of Condensin during Fast Replication in *B. subtilis*. *Current Biology* 24:293-298 <https://doi.org/10.1016/j.CUB.2013.12.049> | PubMed
- Hirano M., Hirano T. (2004) Positive and negative regulation of SMC-DNA interactions by ATP and accessory proteins. *The EMBO journal* 23:2664-2673 <https://doi.org/10.1038/sj.emboj.7600264> | PubMed
- Hutter C.A.J., et al. (2019) The extracellular gate shapes the energy profile of an ABC exporter. *Nature Communications* 10 <https://doi.org/10.1038/s41467-019-09892-6> | PubMed
- Jumper J., et al. (2021) Highly accurate protein structure prediction with AlphaFold. *Nature* 596:583-589 <https://doi.org/10.1038/s41586-021-03819-2> | PubMed
- Kruse A.C., et al. (2013) Activation and allosteric modulation of a muscarinic acetylcholine receptor. *Nature* 504:101-106 <https://doi.org/10.1038/nature12735> | PubMed
- Liu H.W., et al. (2025) The SMC Hinge is a Selective Gate for Obstacle Bypass. *bioRxiv* <https://doi.org/10.1101/2025.03.17.643644>
- Mantes A.D., et al. (2024) Spotiflow: accurate and efficient spot detection for fluorescence microscopy with deep stereographic flow regression. *bioRxiv* <https://doi.org/10.1101/2024.02.01.578426>
- Marko J.F., et al. (2019) DNA-segment-capture model for loop extrusion by structural maintenance of chromosome (SMC) protein complexes. *Nucleic Acids Research* 47:6956-6972 <https://doi.org/10.1093/nar/gkz497> | PubMed
- Minnen A., et al. (2016) Control of SMC Coiled Coil Architecture by the ATPase Heads Facilitates Targeting to Chromosomal ParB/parS and Release onto Flanking DNA. *Cell Reports* 14 <https://doi.org/10.1016/j.celrep.2016.01.066> | PubMed
- Pacesa M., et al. (2025) One-shot design of functional protein binders with BindCraft. *Nature* 1-10 <https://doi.org/10.1038/s41586-025-09429-6> | PubMed
- Rasmussen Søren G.F., et al. (2011) Structure of a nanobody-stabilized active state of the $\beta(2)$ adrenoceptor. *Nature* 469:175-181 <https://doi.org/10.1038/NATURE09648> | PubMed
- Seeger M.A., et al. (2012) Tuning the Drug Efflux Activity of an ABC Transporter in vivo by in vitro Selected DARPins Binders. *PLoS ONE* 7:e37845 <https://doi.org/10.1371/journal.pone.0037845> | PubMed

- Soh Y.M., et al. (2015) Molecular basis for SMC rod formation and its dissolution upon DNA binding. *Molecular cell* **57**:290-303 <https://doi.org/10.1016/J.MOLCEL.2014.11.023> | PubMed
- Stringer C., et al. (2021) Cellpose: a generalist algorithm for cellular segmentation. *Nature Methods* **18**:100-106 <https://doi.org/10.1038/s41592-020-01018-x> | PubMed
- Sullivan N.L., Marquis K.A., Rudner D.Z. (2009) Recruitment of SMC by ParB-parS Organizes the Origin Region and Promotes Efficient Chromosome Segregation. *Cell* **137**:697-707 <https://doi.org/10.1016/J.CELL.2009.04.044> | PubMed
- Vazquez Nunez R., et al. (2021) Gradual opening of SMC arms in prokaryotic condensin. *Cell Reports* **35**:109051 <https://doi.org/10.1016/J.CELREP.2021.109051> | PubMed
- Vazquez Nunez R., Ruiz Avila L.B., Gruber S. (2019) Transient DNA Occupancy of the SMC Interarm Space in Prokaryotic Condensin. *Molecular Cell* **75**:209-223.e6. <https://doi.org/10.1016/j.molcel.2019.05.001> | PubMed
- Wang X., et al. (2014) The SMC condensin complex is required for origin segregation in *Bacillus subtilis*. *Current Biology* **24**:287-292 <https://doi.org/10.1016/j.cub.2013.11.050> | PubMed
- Wang X., et al. (2017) *Bacillus subtilis* SMC complexes juxtapose chromosome arms as they travel from origin to terminus. *Science* **355**:524-527 <https://doi.org/10.1126/science.aai8982> | PubMed
- Yatskevich S., Rhodes J., Nasmyth K. (2019) Organization of Chromosomal DNA by SMC Complexes. *Annual review of genetics* **53**:445-482 <https://doi.org/10.1146/ANNUREV-GENET-112618-043633> | PubMed
- Zimmermann I., et al. (2018) Synthetic single domain antibodies for the conformational trapping of membrane proteins. *eLife* **7** <https://doi.org/10.7554/eLife.34317> | PubMed
- Zimmermann I., et al. (2020) Generation of synthetic nanobodies against delicate proteins. *Nature Protocols* **15**:1707-1741 <https://doi.org/10.1038/s41596-020-0304-x> | PubMed
- Zimmermann L., et al. (2018) A Completely Reimplemented MPI Bioinformatics Toolkit with a New HHpred Server at its Core. *Journal of Molecular Biology* **430**:2237-2243 <https://doi.org/10.1016/j.jmb.2017.12.007> | PubMed

Peer reviews

Reviewer #1 (Public review):

Summary:

Gosselin et al., develop a method to target protein activity using synthetic single-domain nanobodies (sybodies). They screen a library of sybodies using ribosome/ phage display generated against bacillus SMC-ScpAB complex. Specifically, they use an ATP hydrolysis deficient mutant of SMC so as to identify sybodies that will potentially disrupt SMC-ScpAB activity. They next screen their library in vivo, using growth defects in rich media as a read-out for SMC activity perturbation. They identify 14 sybodies that mirror SMC deletion phenotype including defective growth in fast-growth conditions, as well as chromosome segregation defects. The authors use a clever approach by making chimeras between bacillus and *S. pneumoniae* SMC to narrow-down to specific regions within the bacillus SMC coiled-coil that are likely targets of the sybodies. Using ATPase assays, they find that the sybodies either impede DNA-stimulated ATP hydrolysis or hyperactivate ATP hydrolysis (even in the absence of DNA). The authors propose that the sybodies may likely be locking SMC-ScpAB in the "closed" or "open" state via interaction with the specific coiled-coil region on SMC. I have a few comments that the authors should consider:

Major comments:

- (1) Lack of direct in vitro binding measurements:

The authors do not provide measurements of sybody affinities, binding/ unbinding kinetics, stoichiometries with respect to Smc-ScpAB. Additionally, do the sybodies preferentially interact with Smc in ATP/ DNA-bound state? And do the sybodies affect the interaction of ScpAB with SMC?

It is understandable that such measurements for 14 sybodies is challenging, and not essential for this study. Nonetheless, it is informative to have biochemical characterization of sybody interaction with the Smc-ScpAB complex for at least 1-2 candidate sybodies described here.

(2) Many modes of sybody binding to Smc are plausible

The authors provide an elaborate discussion of sybodies locking the Smc-ScpAB complex in open/ closed states. However, in the absence of structural support, the mechanistic inferences may need to be tempered. For example, is it also not possible for the sybodies to bind the inner interface of the coiled-coil, resulting in steric hinderance to coiled-coil interactions. It is also possible that sybody interaction disrupts ScpAB interaction (as data ruling this possibility out has not been provided). Thus, other potential mechanisms would be worth considering/ discussing. In this direction, did AlphaFold reveal any potential insights into putative binding locations?

(3) Sybody expression in vivo

Have the authors estimated sybody expression in vivo? Are they all expressed to similar levels?

(4) Sybodies should phenocopy ATP hydrolysis mutant of Smc

The sybodies were screened against an ATP hydrolysis deficient mutant of Smc, with the rationale that these sybodies would interfere this step of the Smc duty cycle. Does the expression of the sybodies in vivo phenocopy the ATP hydrolysis deficient mutant of Smc? Could the authors consider any phenotypic read-outs that can indicate whether the sybody action results in an smc-null effect or specifically an ATP hydrolysis deficient effect?

Significance:

Overall, this is an impressive study that uses an elegant strategy to find inhibitors of protein activity in vivo. The manuscript is clearly written and the experiments are logical and well-designed. The findings from the study will be significant to the broad field of genome biology, synthetic biology and also SMC biology. Specifically, the coiled coil domain of SMC proteins have been proposed to be of high functional value. The authors have elegantly identified key coiled-coil regions that may be important for function, and parallelly exhibited potential of the use of synthetic sybody/designed binders for inhibition of protein activity.

<https://doi.org/10.7554/eLife.111131.1.sa3>

Reviewer #2 (Public review):

Summary:

Structural Maintenance of Chromosome proteins (SMCs), a family of proteins found in almost all organisms, are organizers of DNA. They accomplish this by a process known as loop extrusion, wherein double-stranded DNA is actively reeled in and extruded into loops. Although SMCs are known to have several DNA binding regions, the exact mechanism by which they facilitate loop extrusion is not understood but is believed to entail large conformational changes. There are currently several models for loop extrusion, including one wherein the coiled coil (CC) arms open, but there is a lack of insightful experimentation and analysis to confirm any of these models. The work presented aims to provide much-needed

new tools to investigate these questions: conformation-selective sybodies (synthetic nanobodies) that are likely to alter the CC opening and closing reactions.

The authors produced, isolated, and expressed sybodies that specifically bound to *Bacillus subtilis* Smc-ScpAB. Using chimeric Smc constructs, where the coiled coils were partly replaced with the corresponding sequences from *Streptococcus pneumoniae*, the authors revealed that the isolated sybodies all targeted the same 4N CC element of the Smc arms. This region is likely disrupted by the sybodies either by stopping the arms from opening (correctly) or forcing them to stay open (enough). Disrupting these functional elements is suggested to cause the Smc-dependent chromosome organization lethal phenotype, implying that arm opening and closing is a key regulatory feature of bacterial Smc-ScpAB.

Significance:

The authors present a new method for trapping bacterial Smc's in certain conformations using synthetic antibodies. Using these antibodies, they have pinpointed the (previously suggested) 4N region of the coiled coils as an essential site for the opening and closing of the Smc coiled coil arms and that hindering these reactions blocks Smc-driven chromosomal organization. The work has important implications for how we might elucidate the mechanism of DNA loop extrusion by SMC complexes.

<https://doi.org/10.7554/eLife.111131.1.sa2>

Reviewer #3 (Public review):

Summary:

Gosselin et al. use the sybody technology to study effects of in vivo inhibition of the *Bacillus subtilis* SMC complex. Smc proteins are central DNA binding elements of several complexes that are vital for chromosome dynamics in almost all organisms. Sybodies are selected from three different libraries of the single domain antibodies, using the "transition state" mutant Smc. They identify 14 such mutant sybodies that are lethal when expressed in vivo, because they prevent proper function of Smc. The authors present evidence suggesting that all obtained sybodies bind to a coiled-coil region close to the Smc "neck", and thereby interfere with the Smc activity cycle, as evidenced by defective ATPase activity when Smc is bound to DNA.

The study is well done and presented and shows that the strategy is very potent in finding a means to quickly turn off a protein's function in vivo, much quicker than depleting the protein.

The authors also draw conclusions on the molecular mode of action of the SMC complex. They provide a number of suggestive experiments, but in my view mostly indirect evidence for such mechanism.

My main criticism is that the authors have used a single - and catalytically trapped form of SMC. They speculate why they only obtain sybodies from one library, and then only identify sybodies that bind to a rather small part of the large Smc protein. While the approach is definitely valuable, it is biased towards sybodies that bind to Smc in a quite special way, it seems. Using wild type Smc would be interesting, to make more robust statements about the action of sybodies potentially binding to different parts of Smc.

Line 105: Alternatively, the other libraries did not produce good binders or these sybodies were 106 not stably expressed in *B. subtilis*. This could be tested using Western blotting - I am assuming sybody antibodies are commercially available. However, this test is not important for the overall study, it would just clarify a minor point.

Fig. 2B: is odd to count Spo0J foci per cells, as it is clear from the images that several origins must be present within the fluorescent foci. I am fine with the "counting" method, as the images show there is a clear segregation defect when sybodies are expressed, I believe the authors should state, though, that this is not a replication block, but failure to segregate origins.

Testing binding sites of sybodies to the SMC complex is done in an indirect manner, by using chimeric Smc constructs. I am surprised why the authors have not used in vitro crosslinking: the authors can purify Smc, and mass spectrometry analyses would identify sites where sybodies are crosslinked to Smc. Again, I am fine with the indirect method, but the authors make quite concrete statements on binding based on non-inhibition of chimeric Smc; I can see alternative explanations why a chimera may not be targeted.

Smc-disrupting sybodies affect the ATPase activity in one of two ways. Again, rather indirect experiments. This leads to the point Revealing Smc arm dynamics through synthetic binders in the discussion. The authors are quite careful in stating that their experiments are suggestive for a certain mode of action of Smc, which is warranted.

In line 245, they state More broadly, the study demonstrates how synthetic binders can trap, stabilize, or block transient conformations of active chromatin-associated machines, providing a powerful means to probe their mechanisms in living cells. This is off course a possible scenario for the use of sybodies, but the study does not really trap Smc in a transient conformation, at least this is not clearly shown.

Overall, it is an interesting study, with a well-presented novel technology, and a limited gain of knowledge on SMC proteins.

Significance:

The work describes the gaining and use of single-binder antibodies (sybodies) to interfere with the function of proteins in bacteria. Using this technology for the SMC complex, the authors demonstrate that they can obtain a significant of binders that target a defined region is SMC and thereby interfere with the ATPase cycle.

The study does not present a strong gain of knowledge of the mode of action of the SMC complex.

<https://doi.org/10.7554/eLife.111131.1.sa1>

Author response:

General Statements

First, we would like to thank the editor at Review Commons for the efficient handling of our manuscript. We also apologize for our delayed response.

We would like to thank all three reviewers for their careful evaluation of our work and their constructive feedback, which will provide a valuable basis for improving the figures and the text, as described below. We expect to be able to complete the revision following the plan described below quickly.

We would like to note that the reviewer reports (Rev. #1 and Rev. #3) made us realize that the manuscript text was misleading on the following point. Although we used the purified ATP hydrolysis-deficient Smc protein for sybody isolation, this does not restrict the selection to a specific conformation. As described in detail in Vazquez-Nunez et al. (Figure 5), this mutant displays the ATP-engaged conformation only in a smaller fraction of complexes (~25% in the presence of ATP and DNA), consistent with prior in vivo observations reported by Diebold-

Durand et al. (Figure 5). Rather than limiting the selection to a particular configuration, our aim was to reduce the prevalence of the predominant rod state in order to broaden the range of conformations represented during sybody selection. Consistent with this interpretation, only a small number of isolated sybodies show strong conformation-specific binding in the presence or absence of ATP/DNA, as observed by ELISA (now included in the manuscript). We will revise the manuscript text accordingly to clarify this point.

Description of the planned revisions

Reviewer #1 (Evidence, reproducibility and clarity):

Gosselin et al., develop a method to target protein activity using synthetic single-domain nanobodies (sybodies). They screen a library of sybodies using ribosome/ phage display generated against bacillus Smc-ScpAB complex. Specifically, they use an ATP hydrolysis deficient mutant of SMC so as to identify sybodies that will potentially disrupt Smc-ScpAB activity. They next screen their library in vivo, using growth defects in rich media as a read-out for Smc activity perturbation. They identify 14 sybodies that mirror smc deletion phenotype including defective growth in fast-growth conditions, as well as chromosome segregation defects. The authors use a clever approach by making chimeras between bacillus and S. pneumoniae Smc to narrow-down to specific regions within the bacillus Smc coiled-coil that are likely targets of the sybodies. Using ATPase assays, they find that the sybodies either impede DNA-stimulated ATP hydrolysis or hyperactivate ATP hydrolysis (even in the absence of DNA). The authors propose that the sybodies may likely be locking Smc-ScpAB in the "closed" or "open" state via interaction with the specific coiled-coil region on Smc. I have a few comments that the authors should consider:

Major comments:

(1) Lack of direct in vitro binding measurements:

The authors do not provide measurements of sybody affinities, binding/ unbinding kinetics, stoichiometries with respect to Smc-ScpAB. Additionally, do the sybodies preferentially interact with Smc in ATP/ DNA-bound state? And, do the sybodies affect the interaction of ScpAB with SMC?

It is understandable that such measurements for 14 sybodies is challenging, and not essential for this study. Nonetheless, it is informative to have biochemical characterization of sybody interaction with the Smc-ScpAB complex for at least 1-2 candidate sybodies described here.

We agree with the reviewer that adding such data would be reassuring and that obtaining solid data using purified components is not easy even for a smaller selection of sybodies. We have data that show direct binding of Smc to sybodies by various methods including ELISA, pull-downs and by biophysical methods (GCI). Initially, we omitted these data from the manuscript as we are convinced that the mapping data obtained with chimeric SMC proteins is more definitive and relevant. During the revision we will incorporate the ELISA data showing direct binding and also indicating a lack of preference for a specific state of Smc.

(2) Many modes of sybody binding to Smc are plausible

The authors provide an elaborate discussion of sybodies locking the Smc-ScpAB complex in open/ closed states. However, in the absence of structural support, the mechanistic inferences may need to be tempered. For example, is it also not possible for the sybodies to bind the inner interface of the coiled-coil, resulting in steric hinderance to coiled-coil interactions. It is also possible that sybody interaction disrupts ScpAB interaction (as data ruling this possibility out has not been provided). Thus, other potential mechanisms

would be worth considering/ discussing. In this direction, did AlphaFold reveal any potential insights into putative binding locations?

We have attempted to map the binding by structure prediction, however, so far, even the latest versions of AlphaFold are not able to clearly delineate the binding interface. Indeed, many ways of binding are possible, including disruption of ScpAB interaction. However, since the main binding site is located on the SMC coiled coils, the later scenario would likely be an indirect consequence of altered coiled coil configuration, consistent with our current interpretation.

(3) Sybody expression in vivo

Have the authors estimated sybody expression in vivo? Are they all expressed to similar levels?

We have tagged selected sybodies with gfp and performed live cell imaging. This showed that they are all roughly equally expressed and that they localize as foci in the cell presumably by binding to Smc complexes loaded onto the chromosome at ParB/parS sites. We will include this data in the revised version of the manuscript.

(4) Sybodies should phenocopy ATP hydrolysis mutant of Smc

The sybodies were screened against an ATP hydrolysis deficient mutant of Smc, with the rationale that these sybodies would interfere this step of the Smc duty cycle. Does the expression of the sybodies in vivo phenocopy the ATP hydrolysis deficient mutant of Smc? Could the authors consider any phenotypic read-outs that can indicate whether the sybody action results in an smc-null effect or specifically an ATP hydrolysis deficient effect?

As eluded to above, we think that our selection gave rise to sybodies that bind various, possibly multiple Smc conformations. Consistent with this idea, the phenotypes are similar to null mutant rather than the ATP-hydrolysis defective EQ mutant, which display even more severe growth phenotypes. We will add the following notes to the text:

“These conditions favour ATP-engaged particles alongside the typically predominant ATP-disengaged rod-shaped state (add Vazquez Nunez et al., 2021).”

“ELISA data confirm that nearly all clones bind Smc-ScpAB; however, their binding shows little or no dependence on the presence of ATP or DNA.”

Minor comments:

(1) It was surprising that no sybodies were found that could target both bacillus and spneu Smc. For example, sybodies targeting the head regions of Smc that might work in a more universal manner. Could the authors comment on the coverage of the sybodies across the protein structure?

It is rather common that sybodies (like antibodies and nanobodies) exhibit strong affinity differences between highly conserved proteins (> 90 % identity). The underlying reasons for such strong discrimination are i) location of less conserved residues primarily at the target protein surface and ii) the large interaction interface between sybody and target which offers multiple vulnerabilities for disturbance, in particular through bulky side chains resulting in steric clashes. Another frequently observed phenomenon is sybody binding to a dominant epitope, which also often applies to nanobodies and antibodies. A great example for this are the dominant epitopes on SARS-CoV-2 RBDs.

(2) Growth curves (Fig. S3) show a large jump in recovery in growth under sybody induction conditions. Could the authors address this observation here and in the text?

We suppose that this recovery represents suppressor mutants and/or (more likely) improved growth in the absence of functional Smc during nutrient limitation (see Gruber et al., 2013 and Wang et al., 2013). We will add this statement to the text.

| (3) L41- Sentence correction: Loop can be removed.

Ah, yes, sorry for this confusing error. Thank you.

| (4) L525 - *bsuSmc 'E'* :extra E can be removed.

To do. Thank you.

| (5) References need to be properly formatted.

To do. Thank you.

| (6) The authors should add in figure legend for Fig 1i) details on representation of the purple region, and explain the grey strokes for orientation of the loop.

To do.

| (7) How many cells were analysed in the cell biological assays? Legends should include these information.

To Be Included.

Reviewer #1 (Significance):

Overall, this is an impressive study that uses an elegant strategy to find inhibitors of protein activity in vivo. The manuscript is clearly written and the experiments are logical and well-designed. The findings from the study will be significant to the broad field of genome biology, synthetic biology and also SMC biology. Specifically, the coiled coil domain of SMC proteins have been proposed to be of high functional value. The authors have elegantly identified key coiled-coil regions that may be important for function, and parallelly exhibited potential of the use of synthetic sybody/designed binders for inhibition of protein activity.

Reviewer #2 (Evidence, reproducibility and clarity):

Review: "Single Domain Antibody Inhibitors Target the Coiled Coil Arms of the Bacillus subtilis SMC complex" by Ophélie Gosselin et al, Review Commons RC-2025-03280
Structural Maintenance of Chromosome proteins (SMCs), a family of proteins found in almost all organisms, are organizers of DNA. They accomplish this by a process known as loop extrusion, wherein double-stranded DNA is actively reeled in and extruded into loops. Although SMCs are known to have several DNA binding regions, the exact mechanism by which they facilitate loop extrusion is not understood but is believed to entail large conformational changes. There are currently several models for loop extrusion, including one wherein the coiled coil (CC) arms open, but there is a lack of insightful experimentation and analysis to confirm any of these models. The work presented aims to provide much-needed new tools to investigate these questions: conformation-selective sybodies (synthetic nanobodies) that are likely to alter the CC opening and closing reactions.

The authors produced, isolated, and expressed sybodies that specifically bound to Bacillus subtilis Smc-ScpAB. Using chimeric Smc constructs, where the coiled coils were partly replaced with the corresponding sequences from Streptococcus pneumoniae, the authors revealed that the isolated sybodies all targeted the same 4N CC element of the Smc arms. This region is likely disrupted by the sybodies either by stopping the arms

from opening (correctly) or forcing them to stay open (enough). Disrupting these functional elements is suggested to cause the Smc-dependent chromosome organization lethal phenotype, implying that arm opening and closing is a key regulatory feature of bacterial Smc-ScpAB.

In summary, the authors present a new method for trapping bacterial Smc's in certain conformations using synthetic antibodies. Using these antibodies, they have pinpointed the (previously suggested) 4N region of the coiled coils as an essential site for the opening and closing of the Smc coiled coil arms and that hindering these reactions blocks Smc-driven chromosomal organization. The work has important implications for how we might elucidate the mechanism of DNA loop extrusion by SMC complexes.

Some specific comments:

Line 75: "likely stabilizing otherwise rare intermediates of the conformational cycle." - sorry, why is that being concluded? Why not stabilizing longer-lived conformations?

We will clarify this statement!

Line 89: Sorry, possibly our lack of understanding: why first ribosome and then phage display?

Ribosome display offers to screen around 10^{12} sybodies per selection round (technically unrestricted library size), while for phage display, the library size is restricted to around 10^9 sybodies due to the fact that production of a phage library requires transformation of the phagemid plasmid into *E. coli*, thereby introducing a diversity bottleneck. This is why the sybody platform starts off with ribosome display. It switches to phage display from round 2 onwards because the output of the initial round of ribosome display is around 10^6 sybodies, which can be easily transferred into the phage display format. Phage display is used to minimize selection biases. For more information, please consult the original sybody paper (PMID: 29792401).

Line 100: Why was only lethality selected? Less severe phenotypes not clear enough?

Yes, colony size is more difficult to score robustly, as the sizes of individual transformant colonies can vary quite widely. The number of isolated sybodies was at the limit of further analysis.

Line 106: Could it be tested somehow if convex and concave library sybodies fold in Bs?

We did not focus on the non-functional sybody candidates and only sybodies of the loop library turned out to cause functional consequences at the cellular level. Notably, we will include gfp-imaging showing that non-lethal sybodies are expressed to similar levels than toxic sybodies. Given the identical scaffold of concave and loop sybodies (they only differ in their CDR3 length), we expect that the concave sybodies fold in the cytoplasm of *B. subtilis*. For the convex sybodies exhibiting a different scaffold, this will be tested.

Line 125: Could Pxyl be repressed by glucose?

To our knowledge and experience, repression by glucose (catabolite repression) does not work well in this context in *B. subtilis*.

Line 131: The SMC replacement strain is a cool experiment and removes a lot of doubts!

Thank you! (we agree).

Line 141: The mapping is good and looks reliable, but looks and feels like a tour de force? Of course, some cryo-EM would have been lovely (lines 228-229 understood, it has been tried!).

Yes, we have made several attempts at structural biology. Unfortunately, Smc-ScpAB is not well suited for cryo-EM in our hands and crystallography with Smc fragments and sybodies did not yield well-diffracting crystals.

Line 179: Mmmh. Do we not assume DNA binding on top of the dimerised heads to open the CC (clamp)?

We will clarify the text here.

Line 187: Having sybodies that presumably keep the CC together (closing) and some that do not allow them to come together correctly (opening) is really cool and probably important going forward.

Thank you!

Figure 1 Ai is not very colour-blind friendly.

We are sorry for this oversight. We will try to make the color scheme more inclusive. Thank you for the notification.

Optional: did the authors see any spontaneous mutations emerge that bypass the lethal phenotype of sybody expression?

No, we did not observe spontaneous mutations suppressing the phenotype, possibly due to the limited number of cell generations observed. We tried to avoid suppressors by limiting growth, but this may indeed be a good future approach for further fine map the binding sites and to obtain insights into the mechanism of inhibition.

Optional: we think it would be nice to try some biochemical experiment with BMOE/cysteine-crosslinked B. subtilis Smc in the mid-region (4N or next to it) of the Smc coiled coils to try to further strengthen the story. Some of the authors are experts in this technique and strains might already exist?

We have indeed tried to study the impact of sybody binding on Smc conformation by cysteine cross-linking. However, we were not convinced by the results and thus prefer not to draw any conclusions from them. We will add a corresponding note to the text.

Reviewer #2 (Significance):

The authors present a new method for trapping bacterial Smc's in certain conformations using synthetic antibodies. Using these antibodies, they have pinpointed the (previously suggested) 4N region of the coiled coils as an essential site for the opening and closing of the Smc coiled coil arms and that hindering these reactions blocks Smc-driven chromosomal organization. The work has important implications for how we might elucidate the mechanism of DNA loop extrusion by SMC complexes.

Thank you!

Reviewer #3 (Evidence, reproducibility and clarity):

Gosselin et al. use the sybody technology to study effects of in vivo inhibition of the Bacillus subtilis SMC complex. Smc proteins are central DNA binding elements of several complexes that are vital for chromosome dynamics in almost all organisms. Sybodies are selected from three different libraries of the single domain antibodies, using the

„transition state" mutant Smc. They identify 14 such mutant sybodies that are lethal when expressed in vivo, because they prevent proper function of Smc. The authors present evidence suggesting that all obtained sybodies bind to a coiled-coil region close to the Smc „neck", and thereby interfere with the Smc activity cycle, as evidenced by defective ATPase activity when Smc is bound to DNA.

The study is well done and presented and shows that the strategy is very potent in finding a means to quickly turn off a protein's function in vivo, much quicker than depleting the protein.

The authors also draw conclusions on the molecular mode of action of the SMC complex. They provide a number of suggestive experiments, but in my view mostly indirect evidence for such mechanism.

My main criticism is that the authors have used a single - and catalytically trapped form of SMC. They speculate why they only obtain sybodies from one library, and then only identify sybodies that bind to a rather small part of the large Smc protein. While the approach is definitely valuable, it is biased towards sybodies that bind to Smc in a quite special way, it seems. Using wild type Smc would be interesting, to make more robust statements about the action of sybodies potentially binding to different parts of Smc.

As explained above, we are quite confident the Smc ATPase mutation did not bias the selection in an obvious way. The surprising bias towards coiled coil binding sites has likely other explanations, as they likely form a preferred epitope recognized by sybodies.

Line 105: Alternatively, the other libraries did not produce good binders or these sybodies were 106 not stably expressed in B. subtilis. This could be tested using Western blotting - I am assuming sybody antibodies are commercially available. However, this test is not important for the overall study, it would just clarify a minor point.

While there are antibody fragments available to augment the size of sybodies (PMID: 40108246), these recognize 3D-epitopes and are thus not suited for Western blotting. We did not follow up on the negative results much, but would like to point out again that there are several biases that likely emerge for the same reason (bias to library, bias to coiled coil binding site). If correct, then likely few other sybodies are effectively lethal in B. subtilis, with the exception of the ones isolated and characterized. We have added this notion to the manuscript. We have also tested the expression of non-lethal sybodies by gfp-tagging and imaging. These results will be included in the revision.

Fig. 2B: it is odd to count Spo0J foci per cells, as it is clear from the images that several origins must be present within the fluorescent foci. I am fine with the „counting" method, as the images show there is a clear segregation defect when sybodies are expressed, I believe the authors should state, though, that this is not a replication block, but failure to segregate origins.

We agree that this is an important point and will add a corresponding comment to the text.

Testing binding sites of sybodies to the SMC complex is done in an indirect manner, by using chimeric Smc constructs. I am surprised why the authors have not used in vitro crosslinking: the authors can purify Smc, and mass spectrometry analyses would identify sites where sybodies are crosslinked to Smc. Again, I am fine with the indirect method, but the authors make quite concrete statements on binding based on non-inhibition of chimeric Smc; I can see alternative explanations why a chimera may not be targeted.

We have made several attempts of testing direct binding with mixed outcomes and decided to not include those results in the light of the stronger and more relevant in vivo mapping.

However, we will add ELISA results and briefly discuss grating coupled interferometry (GCI) data and pull-downs.

Smc-disrupting sybodies affect the ATPase activity in one of two ways. Again, rather indirect experiments. This leads to the point Revealing Smc arm dynamics through synthetic binders in the discussion. The authors are quite careful in stating that their experiments are suggestive for a certain mode of action of Smc, which is warranted.

In line 245, they state More broadly, the study demonstrates how synthetic binders can trap, stabilize, or block transient conformations of active chromatin-associated machines, providing a powerful means to probe their mechanisms in living cells. This is off course a possible scenario for the use of sybodies, but the study does not really trap Smc in a transient conformation, at least this is not clearly shown.

We agree and will carefully rephrase this statement. Thank you.

Overall, it is an interesting study, with a well-presented novel technology, and a limited gain of knowledge on SMC proteins.

We respectfully disagree with the last point, since our unique results highlight the importance of the Smc coiled coils, which are otherwise largely neglected in the SMC literature, likely (at least in part) due the mild effect of single point mutations on coiled coil dynamics.

Reviewer #3 (Significance):

The work describes the gaining and use of single-binder antibodies (sybodies) to interfere with the function of proteins in bacteria. Using this technology for the SMC complex, the authors demonstrate that they can obtain a significant of binders that target a defined region is SMC and thereby interfere with the ATPase cycle.

The study does not present a strong gain of knowledge of the mode of action of the SMC complex.

As pointed out above, we respectfully disagree with this assertion.

Description of analyses that authors prefer not to carry out

As pointed out above, there are a few minor points that we prefer not to experimentally address. In particular, we do not consider it as necessary to determine the expression levels of sybodies which were non-inhibitory. We also wish to note that we attempted to obtain structural additional biochemical data and to that end performed cryo-EM, crystallography and cysteine cross-linking experiments. Unfortunately, we did not obtain sybody complex structures and the cross-linking data were unfortunately not conclusive. We also wish to note that the first author has finished her PhD and left the lab, which limits our capacity to add additional experiments. However, as the reviewers also pointed out, the main conclusions are well supported by the data already.

<https://doi.org/10.7554/eLife.111131.1.sa0>